Incomplete Tariff Pass-Through at the Firm-level: Evidence from the U.S.-China Trade Dispute

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Link to Latest Version

March 23, 2025

Abstract

Recent studies on the U.S.-China trade dispute suggest that the increases in U.S. import tariffs were completely borne by U.S. importers. However, using firm-level data from the U.S. Census, we find that tariff pass-through is incomplete for firms that continue importing the same product from the same country. Large importers experience higher pass-through and account for a greater share of import. Firms that import new products or source from different countries pay higher prices than those maintaining existing relationships. Thus, the observed complete pass-through in prior studies reflects import reallocation toward firms with higher pass-through or more costly new supplier relationships. To explain these patterns, we incorporate a firm-specific import price with two-sided market power into a standard importer model. We show that fixed import costs, an elastic foreign export supply, and the greater bargaining power of large U.S. importers help account for the empirical findings.

Keywords: Trade policy, U.S.-China trade dispute, firm heterogeneity, tariff pass-through.

JEL classification: F10, F30, F40, F51.

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Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2685 (CBDRB-FY24-P2685-R11401). We are extremely grateful to George Alessandria, Yan Bai, Arnaud Costinot, Gaston Chaumont, Jaerim Choi, Jianpeng Deng, Rafael Guntin, Zibin Huang, Chang-Tai Hsieh, Ryan Kim, Sam Kortum, Bingjing Li, Wei Li, Yuhei Miyauchi, Kim Ruhl, Dongling Su, Chang Sun and Xiaodong Zhu for their inspiring comments and suggestions. We thank seminar participants at Yonsei University, Shanghai University of Finance and Economics, the University of Hong Kong, and Xiamen University for their helpful discussions.

1 Introduction

The ongoing U.S.-China trade dispute led to the most abrupt increases in trade barriers in decades. Between 2018 and 2019, import tariffs were imposed on $450 billion in bilateral trade, raising the average tariff rate to 17% (Bekkers and Schroete, 2020). Understanding how import prices respond to import tariffs is crucial for assessing the evolution of international trade and the welfare implications of trade policy.

Recent studies find complete tariff pass-through to U.S. import prices (Amiti et al., 2019b; Cavallo et al., 2021; Fajgelbaum et al., 2020; Jiao et al., 2023), indicating that foreign exporters did not adjust their prices in response to U.S. import tariffs. Therefore, the U.S. imposed import tariffs completely backfired on U.S. importers. This finding is surprising, as previous literature and the presumption suggest that large economies like the U.S. and China should be able to affect prices, leading to incomplete tariff pass-through (Fajgelbaum and Khandelwal, 2022). Notably, existing studies rely primarily on aggregate trade data at the product-country level, which overlooks firm-level variation. Given firms' central role in the importing market and heterogeneity in sourcing strategies, their responses to tariff increases offer valuable insights into the mechanisms underlying complete tariff pass-through.

To understand the drivers of the complete tariff pass-through and the role of firms, we study tariff pass-through both empirically and theoretically. Using confidential data from the U.S. Census Bureau, we decompose pass-through at the firm-product-country level. Our findings show that within stable import relationships, pass-through is incomplete, with significant reallocation across firms. Larger importers experience higher pass-through and account for a larger share of imports. Along the extensive margin, firms importing new products or sourcing from different countries pay higher prices than those maintaining existing supplier relationships. These results suggest that the observed complete tariff pass-through in recent studies is driven by import reallocation toward firms with higher pass-through or more costly new trade relationships. To explain these patterns, we extend a standard importer model to incorporate firm-specific import prices with two-sided market power. Our model demonstrates that fixed import costs, an elastic foreign export supply, and greater bargaining power among larger U.S. importers can account for the empirical findings.

Using a merged dataset from the Longitudinal Firm Trade Transactions Database (LFTTD) and the Longitudinal Business Database (LBD), we decompose the changes in *product-country level* foreign export prices—measured as unit values, or import prices before tariffs—into four components that incorporate firm-level variation: *within-continuing, between-*

continuing, starting, and ending, following the dynamic decomposition method proposed by Melitz and Polanec (2015).

This decomposition highlights how different margins of adjustment to tariff increases shape tariff pass-through at the aggregate level. In terms of intensive margin adjustment, the *within-continuing* component reflects changes in average unit values among firms that continue importing from the same product-country pairs over time. The *between-continuing* component measures changes driven by the reallocation of import shares among firms maintaining ongoing trade relationships. For extensive margin adjustment, the *starting* and *ending* component account for unit value changes due to firms' shift in sourcing strategies by initiating or discontinuing imports from specific product-country pairs. By incorporating firm-level dynamics, this approach provides new insights into macro-micro discrepancies in the pass-through of cost shocks to import prices and the resulting welfare implications.

The decomposition results show that U.S. importers that continue sourcing from the same product-country pairs pay lower prices to foreign exporters following the tariff increase, indicating incomplete pass-through within ongoing trade relationships. Moreover, significant reallocation occurs across product-country pairs, with starting pairs accounting for a larger import share in product-country pairs exposed to higher tariff increase. These findings remain robust across different phases of the U.S.-China trade dispute, alternative tariff measures, and various empirical specifications with different fixed effects.

To further investigate the mechanisms underlying this decomposition, we separately analyze continuing pairs, starting pairs, and ending pairs at the *firm-product-country level*, with a particular focus on the heterogeneous responses of firms of different sizes. First, among firms with continuing pairs, while tariff pass-through is incomplete on average, larger importers experience higher pass-through and account for a greater share of imports following the tariff increase. Second, on average, firms with starting and ending pairs pay higher prices to foreign exporters than those with continuing pairs. Therefore, the reallocation of import shares toward larger firms and the starting margin contribute positively to tariff pass-through, bridging the gap between product-country-level and firm-product-country-level findings.

To rationalize the empirical facts and quantify the contributions of different margins to aggregate tariff pass-through, we develop a new importer model, building on Gopinath and Neiman (2014), Alviarez et al. (2023), and related work. The model incorporates both sourcing strategies across varieties, which stem from the trade-off between fixed import costs and the efficiency gains from importing a broader scope of products, and determinants of import prices and tariff pass-through. Key factors influencing import prices and pass-through include exporters' marginal production costs and two-sided market power, characterized by oligopoly markups and oligopsony markdowns, weighted by importers' relative bargaining power. These markups and markdowns depend on demand and supply elasticities and the relative significance of each trading partner in the other's overall trade. Within this framework, tariff pass-through to import prices operates through four channels: *strategic complementarity channel*, *strategic substitutability channel*, *terms-of-trade channel*, and *sourcing strategy channel*. The first three channels capture the intensive margin adjustments of firms, while the last represents the extensive margin adjustments.

The *strategic complementarity channel* reflects the tendency of foreign exporters to lower their oligopoly markups over marginal production costs in response to declining U.S. demand following a tariff increase. The *strategic substitutability channel* describes how U.S. importers reduce their oligopsony markdowns as they reduce demand for foreign goods after a tariff increase. The *terms-of-trade channel* captures the mechanism by which foreign exporters lower marginal production costs as they reduce output due to scaled-down U.S. demand.

Notably, the *strategic substitutability* and *terms-of-trade* channels are non-degenerate only when exporters' production exhibits decreasing returns to scale, implying a finite foreign export supply elasticity. When exporters operate with constant returns to scale and exhibit elastic export supply, they can easily sell goods to non-U.S. trading partners without altering the marginal cost of production. This renders the *terms-of-trade* channel inactive and prevents importers from adjusting oligopsony markdowns.

Taking the model to the data, we find three key factors that explain the observed tariff pass-through and firms' heterogeneous responses. First, U.S. importers face an elastic foreign export supply, meaning that *strategic substitutability* channel and *terms-of-trade* channel are mostly degenerated. Second, larger importers have greater bargaining power. In this scenario, the relative bargaining power of importers and the oligopoly markup set by exporters jointly determine foreign export prices and the extent of tariff pass-through to the U.S. import prices. When tariffs rise, foreign exporters reduce their markups to mitigate the negative effects of declining demand and to stabilize their market share, leading to incomplete pass-through. However, when larger importers have greater bargaining power, exporters' markups play a smaller role in price determination, leaving exporters with less flexibility to adjust their prices. As a result, larger importers experience greater pass-through. Third, large importers pay higher import prices than smaller importers because they import foreign goods with higher marginal production costs. After a tariff increase, the number of importers declines, particularly among smaller firms that are less able to cover fixed import costs. Since larger firms pay higher prices on average, the exit of smaller firms implies that U.S. firms, on aggregate, face higher import prices after a tariff hike. Thus, the *sourcing strategy* channel helps reconcile the macro-micro discrepancy in tariff pass-through. While we do not directly examine why larger importers face higher marginal production costs from exporters, one possible explanation is that they purchase higher-quality goods, which are inherently more expensive, as suggested by previous literature (see, e.g., Blaum et al. (2019)).

The model highlights that previous literature, which overlooks firm-level variation, overestimates the welfare costs of tariff increases. This overestimation arises from two sources. First, the failure to account for the incomplete tariff pass-through among firms that continue importing from the same product-country pair over time underestimates the gains to U.S. importers. Second, the allocative efficiency gains from increased import reallocation toward more productive firms following a tariff increase are not taken into account.

Moreover, our findings indicate that importing countries with substantial bargaining power or those facing high export supply elasticity are more likely to bear the burden of tariffs when implementing protectionist measures. This occurs because exporters in partner countries may either lack pricing flexibility (under high importers' bargaining power) or possess alternative market access (under high export supply elasticity), limiting importers' ability to pass costs abroad. The U.S.-China trade dispute exemplifies this dynamic.

Related Literature. Our paper contributes to several strands of literature. First, there is a large literature on the pass-through of cost shocks (tariff or exchange rate) into import prices. The common finding is that the pass-through of cost shocks into prices is incomplete.[1] Existing theories explain incomplete pass-through by taking into account factors such as intermediate input costs, currency of invoicing, age of trading relationships, quality upgrading, and price complementarities among exporters, importers, or competitors, broadly defined, adjusting to exchange rate or tariff changes (see, e.g., Amiti et al., 2014, 2019a, 2022; Auer and Schoenle, 2016; Heise, 2024; Ludema and Yu, 2016; Pennings, 2017). However, recent studies focusing on the U.S.-China trade dispute have found complete tariff pass-through (Amiti et al., 2019b; Cavallo et al., 2021; Fajgelbaum et al., 2020; Jiang et al., 2023; Jiao et al., 2023; Tian et al., 2022) using either U.S. or Chinese bilateral trade data.[2] Alviarez

[1] Papers that study exchange rate pass-through include Berman et al. (2012), Auer and Schoenle (2016), Garetto (2016), Chen and Juvenal (2016), Devereux et al. (2017), Freitag and Lein (2023), and De Walque et al. (2023), among many others. Several papers have analyzed the response of import prices to tariff changes Feenstra (1989); Fitzgerald and Haller (2018); Kreinin (1977); Winkelmann and Winkelmann (1998). Other papers analyze the tariff pass-through to consumers Flaaen et al. (2019). See Fajgelbaum and Khandelwal (2022) for a recent review.

[2] Jiao et al. (2023) use data at the firm-product-country level in China, but only focus on one prefecture-level

et al. (2023) instead take into account firm-level variation and specifically focuses on the intensive margin adjustment of firms given fixed trading relationships, showing that ignoring two-sided market power could exaggerate tariff pass-through. Compared to previous literature, our paper offers a new perspective on studying pass-through by considering both extensive margin and intensive margin responses. Also, we newly reveal the importance of the extensive margin, which involves switching import source countries or products in response to cost shocks, in rationalizing complete pass-through at the aggregate level.

Second, our paper contributes a new model to the theoretical work studying importer dynamics (see, e.g., Gopinath and Neiman (2014), Halpern et al. (2015), Ramanarayanan (2017), Blaum (2018), Blaum et al. (2018, 2019), Lyu et al. (2023), Bernard et al. (2018)). We develop a comprehensive model that incorporates various mechanisms with two-sided market power and sourcing strategy adjustments. This model is useful to study the intensive adjustments (import price and quantity) and extensive margin adjustments (sourcing strategy) of U.S. importers, as well as firms' heterogeneous behaviors based on their size. This provides a method to reconcile macro-micro discrepancies in tariff pass-through and the resulting welfare implications.

Third, our paper adds new model-implied estimates to the early literature on firm-specific bargaining power (see, e.g., Pfeffer and Salancik (1978), Porter (1980), Sutton (1991), Dobson et al. (1998), Basker (2007), Mayer and Ottaviano (2008), Ellison and Snyder (2010)). In particular, while previous studies have primarily focused on several specific industries or provided anecdotal evidence to argue that larger firms have higher bargaining power, we use a trade model with heterogeneous tariff pass-through among firms to infer that, across all product-country pairs, larger firms possess greater bargaining power. These model-implied estimates are not only useful for future studies on firm bargaining power, but also prove vital for explaining any pass-through of cost shocks into import or export prices.

Fourth, our paper contributes new empirical evidence to the growing set of papers that study trade policy and global supply chain disruptions. Several papers, among many others, emphasize the negative impact of tariff increases via supply chain disruption on imports (Handley et al., 2023; Monarch and Schmidt-Eisenlohr, 2023) and exports (Handley et al., 2020), as well as the reallocation effect on third countries or non-targeted products (Dang et al., 2023).[3] We provide new facts, showing that after a tariff increase, the formation of

city due to data availability issues.

[3]Other papers emphasize the negative impact of tariff increases (or trade policy uncertainty) via supply chain disruption on unintended consequences for downstream industries (Bown et al., 2021; Erbahar and Zi, 2017), firms' financial market performance (Huang et al., 2023), or input quality (Heise et al., 2024).

new trading relationships is more likely to occur among established products and countries, and less likely to happen among new products or new countries in the short run. These facts potentially suggest that the search cost or fixed import cost is larger for importing new products or sourcing countries.[4]

The rest of the paper is organized as follows. Section 2 decomposes aggregate tariff pass-through using firm-product-country level data and presents new empirical facts. Section 3 describes the benchmark model that incorporates different mechanisms. Section 4 discusses the relative importance of different mechanisms in rationalizing the empirical facts, and re-evaluates the welfare implications of a tariff increase. Section 5 concludes.

2 Empirical Facts

In this section, we demonstrate that although there is (more than) complete pass-through at the aggregate level, the tariff pass-through to U.S. firms is incomplete within a fixed trade relationship. To explain the discrepancy between the aggregate-level and firm-level tariff pass-through, we examine the responses of firms to tariff increases, considering both the intensive and extensive margins.

2.1 Data

The baseline sample is constructed from the imports data from Longitudinal Foreign Trade Transactions Database (LFTTD). LFTTD contains transaction level data on imports (for all import shipments above $2,000 and $250 for certain quota items), which is collected by U.S. Customs and Border Protection via import declaration form (U.S. Customs Forms 7501 and 7533). The Customs forms contain EIN, detailed 10-digit Harmonized System product code, value, quantity, country of origin, date of the transaction, port, mode of transport, and whether the transaction is between related parties.

Furthermore, several papers directly study firms' input sourcing decisions and the trade-off between risk and rewards under supply chain risk (Baldwin and Freeman, 2022; Blaum et al., 2023).

[4] Relatedly, Monarch and Schmidt-Eisenlohr (2023) document that more than 80 percent of U.S. imports occur in preexisting firm-to-firm relationships. Grossman et al. (2024) model that firms renegotiate with initial suppliers or search for replacements in response to unanticipated tariffs, and firms' choices depend upon the match-specific productivity, search cost, and relative price between sourcing countries. Ramanarayanan (2017) models that new importers pay a higher fixed cost than incumbents to source from suppliers.

2.2 Aggregate Tariff Pass-Through and Decomposition

We begin by estimating the tariff pass-through at the product-country level, to align our analysis with existing studies on tariff pass-through. Next, we decompose the changes in unit value at the product-country level to examine the sources of the tariff pass-through. This decomposition allows for attributing the estimated tariff pass-through to various import relationships of U.S. importers, namely imported product-country pairs that survive the tariff increases and those entering and exiting the import market.

2.2.1 Tariff Pass-Through Using Product-Country Level Data

In this section, we validate our sample by estimating the tariff pass-through at the product-country level, following Amiti et al. (2019b). Tariff pass-through refers specifically to the extent to which U.S. import tariffs are transmitted to the prices paid by U.S. importers. Our baseline sample extends that of Amiti et al. (2019b) to cover the period from January 2017 to December 2019.

Specifically, we first construct the price for each product-country market, where the product is defined at HS10 level. The price is measured as the unit value, defined as the ratio of before-duty import value to import quantity. The data of before-duty import value and import quantity are directly from LFTTD. We sum up before-duty import value and import quantity across firms for each product-country pair, then the unit value is defined as

$$p_{jct} \equiv \frac{\sum_i v_{ijct}}{\sum_i m_{ijct}}, \tag{2.1}$$

where v_{ijct} denotes firm i's before-duty import value of product j from country c in year-month t, and m_{ijct} denotes firm i's import quantity of product j from country c in year-month t. This unit value can also be interpreted as the price foreign exporters charge.

We study tariff pass-through to import price by estimating

$$\Delta \ln p_{jct} = \alpha + \beta \Delta \ln(1 + \tau_{jct}) + \eta_{ct} + \delta_j + \epsilon_{jct}, \tag{2.2}$$

where $\Delta \ln p_{jct}$ represents the 12-month change in the unit value, calculated as the difference between the unit value in month $t+12$ and that in month t. The severity of U.S.-China trade dispute is measured by the changes in the applied tariff rates. $\Delta \ln(1 + \tau_{jct})$ is the 12-month change of the applied tariff rate for product-country market jc. The product-country level applied tariffs are directly from LFTTD, measured by the ratio of collected duty to import value for each product-country pair jc[5]. The tariff increases during U.S.-China trade dispute was unexpected by U.S. importers and foreign exporters. As Fajgelbaum et al. (2020) argue, there is little evidence of pre-trends in the affected industries, therefore we could interpret the tariff change as a policy surprise. η_{ct} is the country-time fixed effect, which absorbs the effects of aggregate dynamics, such as the exchange rate fluctuations. δ_j controls for the product fixed effect. Standard errors are clustered at HS-8 digit level. We use 12-month difference in unit value as our baseline analysis to take into account the seasonality issue or the lumpy nature of imports. Moreover, what matters is the difference between U.S.-China trade dispute period and previous period, so it is innocuous to use 12-month difference as the main analysis.

The coefficient of interest is β. If the value of β is negative (positive), it implies that the foreign exporters decrease (increase) prices in response to the tariff increases. On the other hand, if the value is significantly close to zero or insignificant, it indicates that that foreign exporters barely change their prices, and all of the tax burden is passed on to U.S. importers, which is called "complete tariff pass-through" in the literature (see among Amiti et al. 2019b; Cavallo et al. 2021; Fajgelbaum et al. 2020; Jiao et al. 2023). The estimation result is presented in Column (1) of Table 2.1. The estimated β is significantly positive, indicating that foreign export prices (unit value) go up to accommodate the tariff increases. That is to say, there is not only complete tariff pass-through to U.S. importers, but also associated increasing import cost.

In a closely related work, Amiti et al. (2019b) use a monthly sample ranging from January 2017 to December 2018 and estimate the short-run effects over a few months after the tariff increases. In comparison, our estimation results capture medium-run effects of the tariff increases. In the short run, as their estimation suggested, the foreign exporter price is barely responsive due to price stickness. In the medium-run, we could observe changing foreign export price to accommodate the tariff changes. Interestingly, we find increasing foreign export price in response to the tariff changes, i.e., more than complete tariff pass-through.

The tariff pass-through evaluated using product-country level data may be propelled by changes in compositional effects, such as the reallocation of market shares between importing firms with varying import prices or shifts in firms' sourcing strategies. Therefore, we utilize firm-level variation to uncover the origins of the pass-through in the next section.

[5] If there are no transactions for a specific product-country pair in any given month, we leave it as missing, instead of substituting alternative values for the applied tariff.

Table 2.1: **Decomposition of Aggregate Tariff Pass-through**

	(1) Unit value	(2) Average	(3) Between	(4) Starting	(5) Ending	(6) $\frac{x^E_{jct'}}{p_{jct}}$	(7) $\frac{p^E_{jct'}-p^S_{jct'}}{p_{jct}}$	(8) $\frac{x^X_{jct}}{p_{jct}}$	(9) $\frac{p^S_{jct}-p^X_{jct}}{p_{jct}}$
$\Delta \ln(1+\tau_{jct})$	0.84	-0.247	-0.253	1.821	-1.271	0.615	4.525	0.358	-1.872
	(2.23)**	(1.88)*	(4.48)***	(8.77)***	(12.77)***	(1.15)	(8.05)***	(15.76)***	(9.57)***
Obs.	2099000	2099000	2099000	2099000	2099000	2099000	2099000	2099000	2099000
adjusted R^2	0.04	0.02	0	0.06	0.08	0.08	0.07	0.35	0.07

Notes: The table presents baseline regressions (2.2), where we project changes in unit value and its four components on changes in tariff. The sample ranges from January 2017 to December 2019. The difference terms are calculated as 12-month log difference. The "Starting" column refers to the starting component in the decomposition 2.4, while the "Ending" column represents the estimation results for ending component. Columns (1) and (2) present the results for responses of share of starting country-product pairs and differences in import prices between starting and continuing pairs. Columns (3) and (4) are the counterparts for exit component. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

2.2.2 Decomposition of Aggregate Tariff Pass-Through

To better understand the sources of tariff pass-through, we decompose the changes in the unit value p_{jct} at the product-country level into the contributions of surviving, entering, and exiting product-country pairs of firms. Specifically, the unit value p_{jct} at the product-country level can be rewritten as

$$p_{jct} \equiv \frac{\sum_i v_{ijct}}{\sum_i m_{ijct}} = \sum_i \frac{m_{ijct}}{\sum_i m_{ijct}} \frac{v_{ijct}}{m_{ijct}} = \sum_i x_{ijct} p_{ijct}, \tag{2.3}$$

where p_{ijct} denotes the unit value for a firm i importing product j from country c, and $x_{ijct} = \frac{m_{ijct}}{\sum_i m_{ijct}}$ denotes a firm i's import quantity share among all firms that import from the same product-country pair.

Following by Melitz and Polanec (2015), the change in unit value at country-product level from period t to period t' can be decomposed as the following:

$$\begin{aligned}\frac{p_{jct'} - p_{jct}}{p_{jct}} = &\underbrace{\frac{\bar{p}^S_{jct'} - \bar{p}^S_{jct}}{p_{jct}}}_{\text{Within Continuing}} + \underbrace{\frac{\sum\limits_{i \in S_{jc}} (x_{ijct'} - \bar{x}^S_{jct'})(p_{ijct'} - \bar{p}^S_{jct'}) - \sum\limits_{i \in S_{jc}} (x_{ijct} - \bar{x}^S_{jct})(p_{ijct} - \bar{p}^S_{jct})}{p_{jct}}}_{\text{Between Continuing}} \\ &+ \underbrace{x^E_{jct'} \frac{\bar{p}^E_{jct'} - \bar{p}^S_{jct'}}{p_{jct}}}_{\text{Starting}} + \underbrace{x^X_{jct} \frac{\bar{p}^S_{jct} - \bar{p}^X_{jct}}{p_{jct}}}_{\text{Ending}}.\end{aligned} \tag{2.4}$$

We explain each of the four components in order.

Within Continuing The first component stands for the changes in the unweighted average unit value across firms that continue importing from a product-country pair from period t to period t'. Let S_{jc} be a set of firms i that import product j from country c in both period t and t'. Then $\bar{p}^S_{jct}$ is the average of unit value in the set S in period t, given as

$$\bar{p}^S_{jct} \equiv \sum_{i \in S_{jc}} \frac{p_{ijct}}{n^S_t},$$

where n^S_t is the size of set S. Intuitively, this term measures the aggregate price changes resulting from shifts in the distribution of unit values among firms that continue to import the same product-country pair over time.

Between Continuing The second component captures the aggregate price changes induced by the reallocation of import shares among firms with continuing trade relationship, i.e., among firms that continue importing from a product-country pair from period t to period t'. Using a similar notation as above,

$$\bar{x}^S_{jct} \equiv \sum_{i \in S_{jc}} \frac{x_{ijct}}{n^S_t}$$

is the unweighted average of import quantity share of firms with a continuing relationship in period t. This term can be understood as the covariance change between market share and the prices.

Starting The third component captures the unit value of the firms with a new trade relationship, that is, those who start importing a new product j or from a different country c. More precisely, let E_{jc} be the set of firms i with zero imports in period t but positive imports in period t' within the product-country pair (j, c). Then

$$x^E_{jct'} \equiv \sum_{i \in E_{jc}} x_{ijct'}$$

measures total import quantity share of firms that start importing product j from country c. Multiplied with the average unit value

$$\bar{p}^E_{jct'} \equiv \sum_{i \in E_{jc}} \frac{p_{ijct'}}{n^E_{t'}}$$

across starting firms relative to average $\bar{p}^{S}_{jct'}$ across continuing firms, this term represents the changes accounted for by the starting pairs of firms.

Ending The last component measures the changes accounted by the ending pairs of firms that end a trade relationship, i.e., that stop importing product j from country c. Ending pairs of firms are those with positive imports of country-product variety in period t' but zero imports in period t, consisting of set X_{jc}. Here, we employ a similar notation as

$$x^{X}_{jct} \equiv \sum_{i \in X_{jc}} x_{ijct} \qquad \bar{p}^{X}_{jct} \equiv \sum_{i \in X_{jc}} \frac{p_{ijct}}{n^{X}_{t}}.$$

With these four components, we revisit the regression of Equation (2.2). That is, we rerun the regression by replacing the left-hand-side, the aggregate unit value change, with one of its four decomposed components. For example, the empirical specification when we replace with the within-continuing component is

$$\frac{\bar{p}^{S}_{jct'} - \bar{p}^{S}_{jct}}{p_{jct}} = \alpha + \beta \Delta \ln(1 + \tau_{jct}) + \eta_{ct} + \delta_j + \epsilon_{jct}. \tag{2.5}$$

In Table 2.1, Column (2) shows that for firms with continuing trading relationships, unit values decrease in response to higher tariffs. This indicates that tariff pass-through is *incomplete* on average within stable trade relationships. That is, foreign exporters with stable U.S. partners lower their export prices to offset tariff increases. Column (3) reports how the covariance change between market share and unit value of continuing pairs responds to tariff increases. The significant negative estimated coefficient indicates that with tariff increases, continuing pairs with smaller unit price compared to the average account for a larger market share. However, the adjusted R^2 is zero. When we use firm-product-country level regressions, we find continuing pairs with higher unit price compared to the average account for a larger market share.

The results of continuing pairs suggest that there is uneven sharing of tariff burdens across U.S. importers that is masked when using the aggregate data and find the complete pass-through on average in the short-run and more than complete tariff pass-through in the medium run. Besides the different level of tariff pass-through across continuing pairs, those starting and ending pairs also contribute to the aggregate tariff path-through. From the decomposition in equation 2.4, the starting (ending) components responses to increased tariffs could be attributed to either changes in share of starting (ending) trade relationships, $\frac{x^E_{jct'}}{p_{jct}}$ $\left(\frac{x^X_{jct}}{p_{jct}}\right)$, or the differences in import prices between starting (ending) and continuing pairs, $\frac{p^E_{jct'}-p^S_{jct'}}{p_{jct}}$ $\left(\frac{p^S_{jct}-p^X_{jct}}{p_{jct}}\right)$.

2.2.3 Further Decomposing Starting and Ending Components

To differentiate, we replace the dependent variables in Equation (2.2) with either starting (ending) shares or price differences relative to continuing pairs. For example, the empirical specification of the starting share is given as

$$\frac{x^E_{jct'}}{p_{jct}} = \alpha + \beta\Delta\ln(1+\tau_{jct}) + \eta_{ct} + \delta_j + \epsilon_{jct}. \tag{2.6}$$

The last four columns in Table 2.1 presents the estimation results for equation 2.6. An increase in tariffs is associated with higher market share of firms that start and end a trade relationship, as suggested by the results in Columns (6) and (8), respectively. Furthermore, we observe an increase in the unit value difference between starting pairs and continuing pairs (Column 7). On the other hand, there is a decrease in the unit value difference between continuing pairs and ending pairs (Column 9). We need to further compare the unit values between continuing pairs and entering (exiting) pairs to obtain how their unit values respond to tariff increases. We continue this discussion in Section 2.4.

Combining with the decomposition of the four components mentioned above, this decomposition yields two key findings. Firstly, given the trading relationship between firms, foreign exporters charge lower export prices, resulting in an incomplete tariff pass-through to U.S. importers. Secondly, there is a significant reallocation across firms with different trade relationships.

To summarize, in Table 2.1, we utilize the firm-level variation within product-country pairs to identify the differential effect of tariff changes to each decomposed component. This approach allows us to disentangle the distinct impacts of tariffs on various margins of price adjustment. However, to fully comprehend the relative importance of different margins to the aggregate tariff pass-through and the resulting welfare implications, the structural model in Section 3 is indispensable.

Table 2.2: Robustness Using Scheduled Tariffs

	(1) Unit value	(2) Within	(3) Between	(4) Starting	(5) Ending	(6) $\frac{x^E_{jct'}}{p_{jct}}$	(7) $\frac{p^E_{jct'}-p^S_{jct'}}{p_{jct}}$	(8) $\frac{x^X_{jct}}{p_{jct}}$	(9) $\frac{p^S_{jct}-p^X_{jct}}{p_{jct}}$
$\Delta \ln(1+\tau^{scheduled}_{jct})$	0.13	-0.108	0.017	0.155	-0.029	-0.591	0.285	0.042	0.039
	(1.96)**	(2.45)**	(0.77)	(4.20)***	(1.32)	(1.37)	(2.66)***	(8.62)***	(0.77)
Obs.	230000	230000	230000	230000	230000	230000	230000	230000	230000
adjusted R^2	0.12	0.04	0.01	0.22	0.28	0.18	0.17	0.53	0.2

Notes: The table presents robustness check for the baseline results in Table 2.1, using scheduled tariffs. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. * significant at 10%; ** significant at 5%; *** significant at 1%.

2.2.4 Robustness

In this section, we demonstrate that our product-country-level empirical findings remain robust across alternative measures of key variables, different sample selections, and various fixed-effect specifications.

Scheduled tariff In line with the literature, including our direct counterpart Amiti et al. (2019b), we use applied tariffs to construct the baseline measure of tariff increases[6]. As a robustness check, we construct an alternative measure of tariff increases based on the scheduled tariff data from Fajgelbaum et al. (2020).[7] This dataset provides monthly, country-product-level information, including the effective month of tariff changes in 2018, as well as the maximum and scaled tariff rates for each year. The detailed process for constructing the scheduled tariff measure is described in Appendix A.2. This robustness check addresses the potential endogeneity concern associated with applied tariffs, as scheduled tariffs are predetermined and not subject to contemporaneous economic conditions.

Table 2.2 presents the estimation results using the scheduled tariff data. The findings confirm the presence of the gap between complete and incomplete tariff pass-through, as shown in Columns (1) and (2). Specifically, foreign exporters, on average, increase their export prices in response to tariff increases, as shown in Column (1). Moreover, the baseline results for incomplete tariff pass-through remain robust within stable trade relationships (Column (2)). Besides, the estimation results for both the entry and exit components indicate observable reallocations across product-country pairs, reflecting shifts in import strategies.

[6] This choice is driven by the substantial loss of observations that would occur if scheduled tariffs were used instead.

[7] We do not utilize the statutory tariff data from Fajgelbaum et al. (2020) due to its sparse time-series coverage.

Table 2.3: Robustness with Alternative Fixed Effects

	(1) Unit value	(2) Within	(3) Between	(4) Starting	(5) Ending	(6) $\frac{x^E_{jct'}}{p_{jct}}$	(7) $\frac{p^E_{jct'}-p^S_{jct'}}{p_{jct}}$	(8) $\frac{x^X_{jct}}{p_{jct}}$	(9) $\frac{p^S_{jct}-p^X_{jct}}{p_{jct}}$
A. Alternative Fixed Effect 1									
$\Delta \ln(1+\tau_{jct})$	0.538	-0.263	-0.281	0.941	-0.658	0.245	4.234	0.227	-1.565
	(1.38)	(1.92)*	(4.94)***	(4.36)***	(6.61)***	(0.37)	(7.24)***	(9.36)***	(8.35)***
Obs.	2096000	2096000	2096000	2096000	2096000	2096000	2096000	2096000	2096000
adjusted R^2	0.05	0.02	-0.01	0.08	0.1	0.09	0.08	0.36	0.1
B. Alternative Fixed Effect 2									
$\Delta \ln(1+\tau_{jct})$	0.692	-0.359	-0.258	1.328	-0.812	0.773	5.097	0.32	-1.838
	(1.70)*	(2.55)**	(4.36)***	(5.73)***	(7.06)***	(0.97)	(8.15)***	(10.74)***	(8.41)***
Obs.	2099000	2099000	2099000	2099000	2099000	2099000	2099000	2099000	2099000
adjusted R^2	0.03	0.01	0	0.02	0.03	0.02	0.04	0.25	0.04
C. Alternative Fixed Effect 3									
$\Delta \ln(1+\tau_{jct})$	0.75	-0.271	-0.228	1.301	-0.794	0.782	4.569	0.333	-1.875
	(2.02)**	(2.05)**	(4.11)***	(6.16)***	(7.52)***	(1.13)	(7.99)***	(12.10)***	(9.23)***
Obs.	2100000	2100000	2100000	2100000	2100000	2100000	2100000	2100000	2100000
adjusted R^2	0.03	0.01	0	0.03	0.03	0.02	0.04	0.25	0.04
D. Alternative Fixed Effect 4									
$\Delta \ln(1+\tau_{jct})$	0.538	-0.263	-0.281	0.941	-0.658	0.245	4.234	0.227	-1.565
	(1.38)	(1.92)*	(4.94)***	(4.36)***	(6.61)***	(0.37)	(7.24)***	(9.36)***	(8.35)***
Obs.	2096000	2096000	2096000	2096000	2096000	2E+06	2096000	2096000	2096000
adjusted R^2	0.05	0.02	-0.01	0.08	0.1	0.09	0.08	0.36	0.1

Notes: The table presents robustness check for the baseline results in Table 2.1 with alternative fixed effects. The difference terms are calculated as 12-month log difference. Panel A controls for country-time and country-sector fixed effects. Panel B controls for country-time and sector-time fixed effects. Panel C controls for country-time and sector fixed effects. Panel D controls for country-time, sector, and country-sector fixed effects. The sector is defined at the HS4 level. Standard errors are clustered at HS-8 digit level. * significant at 10%; ** significant at 5%; *** significant at 1%.

Alternative Fixed Effects Our baseline results control for country-time and product fixed effects, which absorb time-varying country-level aggregate shocks and time-invariant differences across products. Motivated by Fajgelbaum et al. (2020) and Jiao et al. (2023), we check the robustness of our results to alternative sets of fixed effects. The results are summarized in Table 2.3, indicating that our findings of incomplete tariff pass-through and reallocation across pairs remain robust.[8]

General Tariff Increases v.s. Tariffs on Chinese Imported Goods Although the Trump administration initiated an investigation into imposing tariffs on China as early as March 22, 2018, the first round of U.S. import tariffs on 34 billion worth of Chinese goods did not take effect until July 8, 2018. Prior to July, the tariff increases were universally directed towards all U.S. trading partners. For instance, tariffs were imposed on solar panels and washing machines on January 22, and subsequently, the Trump administration announced

[8]Dealing with endogeneity of tariffs is, of course, a concern in analyses of the effects of trade policy (Goldberg and Pavcnik, 2016). In the case of the trade war, this research has demonstrated that, conditional on the fixed effects that enter in the controls, the tariff changes were uncorrelated with previous price and import trends across products (See Cavallo et al. 2021, Fajgelbaum et al. 2020, among others).

Table 2.4: **Robustness Differentiating Chinese Specific Tariffs**

	(1) Unit value	(2) Within	(3) Between	(4) Starting	(5) Ending	(6) $\frac{x^E_{jct'}}{p_{jct}}$	(7) $\frac{p^E_{jct'}-p^S_{jct'}}{p_{jct}}$	(8) $\frac{x^X_{jct}}{p_{jct}}$	(9) $\frac{p^S_{jct}-p^X_{jct}}{p_{jct}}$
A. General Tariff Increases									
$\Delta \ln(1+\tau_{jct})$	0.965	-0.707	-0.361	2.12	-1.527	0.067	6.886	0.319	-2.124
	(1.47)	(2.67)***	(3.22)***	(5.09)***	(7.14)***	(0.11)	(6.30)***	(8.56)***	(5.32)***
Obs.	516000	516000	516000	516000	516000	516000	516000	516000	516000
adjusted R^2	0.05	0.02	-0.01	0.06	0.08	0.09	0.07	0.35	0.08
B. Tariffs on Chinese Imported Goods									
$\Delta \ln(1+\tau_{jct})$	0.974	-0.012	-0.236	1.739	-1.18	0.63	4.006	0.364	-1.767
	(2.63)***	(0.09)	(3.86)***	(8.35)***	(10.66)***	(0.98)	(6.93)***	(14.11)***	(8.15)***
Obs.	1583000	1583000	1583000	1583000	1583000	1583000	1583000	1583000	1583000
adjusted R^2	0.04	0.02	0	0.06	0.08	0.07	0.07	0.35	0.07

Notes: The table presents robustness check for the baseline results in Table 2.1 using alternative data samples. The difference terms are calculated as 12-month log difference. Compared to the baseline results, Panel A applies sample ranging from January 2017 to June 2018, while the sample of Panel B is from July 2018 to December 2019. * significant at 10%; ** significant at 5%; *** significant at 1%.

tariffs on steel and aluminum imports. After July, U.S. import tariffs specifically targeted China. Therefore, we designate July 2018 as a cutoff and divide our baseline sample into two sub-periods.

We perform the same baseline regressions for both sub-periods to assess the robustness of our empirical findings. As depicted in Panel A and B of Table 2.4, whether during the period of general tariff increases or the period of China-specific tariffs, our baseline findings remain consistent. Interestingly, the incomplete pass-through for continuing pairs that keep importing from China is insignificant. However, at the aggregate level, there is more than complete tariff pass-through. This is driven by the adjustment of starting pairs. Therefore, the adjustments of starting pairs in both periods of the tariff dispute are important for understanding the aggregate-level tariff pass-through.

Pre-dispute Period v.s. Dispute Period. We extend the baseline sample to include data from January 2015 to December 2019. This extended sample includes periods without significant tariff changes, allowing us to compare the effects of tariff increases with grace periods in the affected product-country markets. Using this alternative sample, we apply the same specification as in Equation (2.1) and find that the baseline correlations between the decomposed components and increased tariffs remain robust, as shown in Table 2.5.

To summarize, we find the more-than-complete aggregate tariff pass-through coexists with the incomplete tariff pass-through of continuing pairs, due to the market share reallocation across product-country pairs, and the import price differences among firms. To better understand the sources of this incomplete pass-through and the reallocation across

Table 2.5: Robustness with Pre-dispute Period

	(1) Unit value	(2) Within	(3) Between	(4) Starting	(5) Ending	(6) $\frac{s^E_{jct'}}{p_{jct}}$	(7) $\frac{p^E_{jct'}-p^S_{jct'}}{p_{jct}}$	(8) $\frac{s^X_{jct}}{p_{jct}}$	(9) $\frac{p^S_{jct}-p^X_{jct}}{p_{jct}}$
$\Delta\ln(1+\tau_{jct})$	0.724	-0.246	-0.365	1.888	-1.497	0.612	5.035	0.279	-1.888
	(1.82)*	(2.13)**	(7.95)***	(8.92)***	(16.20)***	(0.93)	(9.47)***	(14.40)***	(11.56)***
Obs.	4019000	4019000	4019000	4019000	4019000	4019000	4019000	4019000	4019000
adjusted R^2	0.04	0.01	0	0.06	0.08	0	0.07	0.34	0.07

Notes: The table presents the corresponding regressions results following (2.1) using an alternative sample ranging from January 2015 to December 2019. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

product-country pairs, we incorporate firm-level variation into our analysis in the subsequent sections. Specifically, we examine how tariff pass-through varies by firm size and explore how starting and ending pairs respond to tariff changes in comparison to continuing pairs.

2.3 Understanding Continuing Pairs

In this section, we focus on continuing pairs that maintain imports from a specific product-country market despite tariff increases. We analyze how foreign export prices and market shares respond to tariff increases at the firm-product-country level, and how these responses vary by firm size.

First, we estimate the average effects of tariff increases at the firm-product-country level:

$$\Delta y_{ijct} = \alpha + \beta \Delta \ln(1+\tau_{jct}) + \eta_{ct} + \delta_j + \epsilon_{ijct} \tag{2.7}$$

where Δy_{ijct} denotes the change in (i) log of unit value of firm i in product-country market jc, $p_{ijct} \equiv V_{ijct}/Q_{ijct}$; (ii) import quantity share within a product-country market s_{ijct}; or (iii) log of after-duty import price $p_{ijct}(1+\tau_{ijct})$. We control for country-year-month (η_{ct}) and product fixed effects (δ_j). The coefficient of interest, β, captures the responses to the tariff increases.

The estimation results are in Panel A of Table 2.6. When tariffs increase, the unit values of continuing firm-product-country pairs decrease on average (Column 1), confirming the finding from Section 2.2 that tariff pass-through to U.S. importers is incomplete. However, the after-duty price still rises (Column 3), indicating that the tax burden is shared between foreign exporters and U.S. importers. Moreover, we observe that the import quantity share increases with tariff changes (Column 2), suggesting that tariff increases drive a redistribution of import shares toward product-country pairs experiencing larger tariff increases.

Next, we study how the responses of continuing pairs depend on firm size. We measure

Table 2.6: Responses of Continuing Pairs

	(1) Δ ln(unit value)	(2) Δimport quantity share	(3) Δ ln(import price)
A. Average Effect			
$\Delta\ln(1+\tau_{jct})$	-0.178	0.05	0.805
	(8.00)***	(16.82)***	(32.88)***
Obs.	7571000	7571000	7571000
FEs	Yes	Yes	Yes
adjusted R^2	0.02	0.01	0.02
B. Heterogeneous effects			
$\Delta\ln(1+\tau_{jct})$	-0.305	-0.07	0.666
	(5.71)***	(5.90)***	(11.98)***
$\ln(emp)_{i,2017}$	-0.001	0	-0.001
	(4.82)***	(1.58)	(4.23)***
$\Delta\ln(1+\tau_{jct})\times\ln(emp)_{i,2017}$	0.03	0.011	0.025
	(4.64)***	(8.29)***	(3.81)***
Obs.	2723000	2723000	2723000
FEs	Yes	Yes	Yes
adjusted R^2	0.02	0.01	0.02

Notes: The table presents the estimation results of estimating equation (2.7) and (2.8). The difference terms are calculated as 12-month log difference. Panel A estimates the average effects of tariff increases, while Panel B shows the corresponding heterogeneous effects. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

firm size as the log of employment in 2017, $\ln(emp)_{i,2017}$, before the trade dispute. We extend the estimation specification by

$$\Delta y_{ijct} = \alpha + \beta_1\Delta\ln(1+\tau_{jct}) + \beta_2\ln(emp)_{i,2017} + \beta_3\Delta\ln(1+\tau_{jct})\times\ln(emp)_{i,2017} + \eta_{ct} + \delta_j + \epsilon_{ijct}. \tag{2.8}$$

The coefficient of interest is β_3, which measures how heterogeneous continuing pairs respond to increased tariffs. The estimation results are shown in Panel B of Table 2.6. Continuing pairs of larger firms have higher unit values, post-tariff import prices, and a larger import quantity share when tariffs go up. In other words, larger firms have higher tariff pass-through and constitute a larger import market share after a tariff increase.

We then replace the employment with a firm's import quantity share in each specific product-country market (jc), defined as $x_{ijc,2017} = q_{ijc,2017}/\sum_i q_{ijc,2017}$. Furthermore, we classify firm-product-country pairs into four groups based on their 2017 import quantity shares to better understand how firm size heterogeneity in each market influences the effects of tariff increases. For instance, $\mathcal{I}^{x2}_{ijc,2017}$ equals 1 if the import quantity share falls within the second quartile of the entire sample.

Table 2.7 summarizes the results. Our baseline findings regarding the response of continuing pairs remain robust when alternative measures of firm size are used, as shown in Panel A. Furthermore, the heterogeneous effects of firm size are primarily driven by firms in the

Table 2.7: **Responses of Continuing Pairs: Alternative Heterogeneity Measure**

	(1)	(2)	(3)
	Δ ln(unit value)	Δimport quantity share	Δ ln(import price)
A. Average Effects			
$\Delta \ln(1+\tau_{jct})$	-0.228	0.053	0.752
	(11.93)***	(16.36)***	(34.46)***
$x_{ijc,2017}$	0.043	-0.056	0.043
	(19.01)***	(79.49)***	(18.93)***
$\Delta \ln(1+\tau_{jct}) \times x_{ijc,2017}$	0.368	-0.021	0.396
	(7.53)***	(1.08)	(8.13)***
Obs.	7162000	7162000	7162000
adjusted R^2	0	0.01	0.01
B. Heterogeneous effects			
$\Delta \ln(1+\tau_{jct})$	-0.325	0.038	0.658
	(5.54)***	(10.91)***	(10.01)***
$\mathcal{I}^{x2}_{ijc,2017}$	0.043	-0.056	0.043
	0.029	0	0.029
	(8.67)***	(1.39)	(8.28)***
$\mathcal{I}^{x3}_{ijc,2017}$	0.051	0.001	0.05
	(15.07)***	(5.01)***	(14.47)***
$\mathcal{I}^{x4}_{ijc,2017}$	0.071	-0.012	0.07
	(20.08)***	(36.29)***	(19.45)***
$\Delta \ln(1+\tau_{jct}) \times \mathcal{I}^{x2}_{ijc,2017}$	0.105	-0.023	0.121
	(1.17)	(7.85)***	(1.25)
$\Delta \ln(1+\tau_{jct}) \times \mathcal{I}^{x3}_{ijc,2017}$	0.152	-0.008	0.157
	(1.78)*	(2.63)***	(1.68)*
$\Delta \ln(1+\tau_{jct}) \times \mathcal{I}^{x4}_{ijc,2017}$	0.214	0.067	0.199
	(2.64)***	(12.08)***	(2.24)**
Obs.	7571000	7571000	7571000
adjusted R^2	0.02	0.01	0.03

Notes: Panel A presents the estimation results of estimating equation (2.8) using import quantity share in 2017 as the measure of firm heterogeneity. Panel B further groups firm-product-country pairs into four groups based on their import quantity share in 2017. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

fourth quartile of import quantity shares.

2.3.1 Understanding Mechanisms Behind

The above results suggest that larger importers experience more complete tariff pass-through. What could be the possible explanations for this? We test each one as discussed below.

Is the heterogeneity in firms' tariff pass-throughs driven by exporter switching? There are two possible scenarios. First, foreign exporters trading with larger U.S. importers reduce their prices by less following a tariff increase. Second, foreign exporters that charge low prices exit the market after tariff increase, and U.S. importers, especially large ones, may switch to foreign exporters that charge higher prices after a tariff increase, even if these foreign exporters do not alter their prices. To further investigate these possibilities, we include exporter information in our control variables. Specifically, among all continuing pairs, for importer i who imports product j from country c in period t, we estimate:

$$\Delta y_{ijct} = \alpha+\beta_1\Delta\ln(1+\tau_{jct})+\beta_2\ln(emp)_{i,2017}+\beta_3\Delta\ln(1+\tau_{jct})\times\ln(emp)_{i,2017}+I_{ijct'}+\eta_{ct}+\delta_j+\epsilon_{ijct}, \tag{2.9}$$

where $I_{ijct'}$ is a dummy variable equal to one if the number of suppliers or the specific suppliers of importer i for product-country pair jc change in the next period t'. The results will be available soon.

Is the heterogeneity in firms' tariff pass-throughs driven by global value chain? It is possible that larger U.S. importers are more likely to import goods from their trading partners that belong to related parties. Therefore, foreign exporters may not need to reduce prices to mitigate the potential decrease in import demand after a tariff increase. Using the same estimation specified in equation (2.9), we replace $I_{ijct'}$ with exports-to-imports$_{ijct}$ to ascertain if this is the case. The results will be available soon.

Is the heterogeneity in firms' tariff pass-throughs driven by variety switching? It is possible that larger U.S. importers are more likely to import more expensive varieties that could be of higher quality or have other more attractive attributes after a tariff increase. This could contribute to the observed heterogeneous tariff pass-through among firms. In particular, Rauch (1999) constructs product differentiation index and shows that some products are more standardized while others are more differentiated and tend to have more varieties. To this end, we have controlled for product fixed effect to rule out this possibility. To further alleviate any concerns, we estimate the following equation:

$$\Delta y_{ijct} = \alpha + \beta_1\Delta\ln(1+\tau_{jct})\times\ln(emp)i, 2017\times I_j + ... + \eta_{ct} + \epsilon_{ijct}, \tag{2.10}$$

where I_j denotes the product differentiation index from Rauch (1999). For standardized products with limited variations within a HS10 category, we expect to see limited variety switching. Therefore, if heterogeneity in firms' tariff pass-through is more significant among standardized products, we can rule out the variety switching explanation. The results will be available soon.

In summary, the finding that larger importers face more tariff pass-throughs appears to be driven not by the switching of varieties or exporters, but by something that depends on or correlates with importer size. As detailed later, we use a standard model with various mechanisms to infer that larger firms have higher bargaining power, which helps explain this finding.

2.4 Understanding Starting and Ending Pairs

Building on our discussion of starting and ending pairs in Section 2.2.3, we expand the analysis by incorporating firm-level variation to examine changes in firms' import strategies. Specifically, we investigate the probability of formation and dissolution of product-country pairs, the price difference and market share reallocation of starting/ending pairs compared to continuing pairs, as well as the differences among firms of varying sizes.

2.4.1 Fewer Starting Pairs, More Ending Pairs

This section examines how tariff changes affects reallocation across firms along the extensive margin. Specifically, whether firms maintain existing import relationships or shift to new product-country markets. We further analyze whether changes along the extensive margin stem from adjustments in imported product categories or shifts in sourcing countries. Lastly, we investigate how these firm-level responses vary with firm heterogeneity, offering insights into the underlying mechanisms.

If a firm has a starting trade relationship, it could be that the firm is importing a new product or sourcing from a different country, or both. On the other hand, a firm can terminate a trading relationship by stopping importing a product, stopping importing from a country, or both. Let $1_{ijct}^{\text{new c-j}} = 1$ denote the firm starts a trading relationship, and let $1_{ijct}^{\text{exit c-j}} = 1$ denote the firm ends a trading relationship. To study whether firms changes in extensive margin in response to tariff increases, we estimate a linear probability model:

$$\mathcal{I}_{ijct} = \alpha + \beta \Delta \ln(1 + \tau_{jct}) + \eta_{ct} + \delta_j + \epsilon_{ijct} \tag{2.11}$$

where the dependent variables are $\mathcal{I}_{ijct} \in \{1_{ijct}^{\text{new c-j}}, 1_{ijct}^{\text{end c-j}}\}$. The estimated parameter of interest is β, which captures how firms' likelihood to create (quit) country-product pairs responses to tariff increases. The results are shown in Panel A of Table 2.8. On average, when tariffs increase, firms are less likely to establish new product-country pairs and more likely to terminate existing import relationships.

Table 2.8: **Probability of Starting and Ending a Trading Relationship**

	$\mathbb{1}^{\text{new c-j}}_{ijct}$	$\mathbb{1}^{\text{exit c-j}}_{ijct}$
A. Average Effects		
$\Delta \ln(1+\tau_{jct})$	-0.173	0.315
	(15.37)***	(27.40)***
Obs.	26110000	26110000
adjusted R^2	0.1	0.04
B. Heterogeneous Effects		
$\Delta \ln(1+\tau_{jct})$	-0.077	0.691
	(2.73)***	(25.31)***
$\ln(emp)_{i,2017}$	-0.006	-0.006
	(24.98)***	(27.93)***
$\Delta \ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	-0.009	-0.057
	(2.71)***	(18.52)***
Obs.	7833000	7833000
adjusted R^2	0.11	0.04

Notes: Panel A and B presents the estimation results of estimating equation (2.11) and (2.12), respectively. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

We further introduce firm heterogeneity into the linear probability model to study how extensive margin responses depend on firms' size. The estimation strategy is given by:

$$\mathcal{I}_{ijct} = \alpha + \beta_1 \Delta \ln(1+\tau_{jct}) + \beta_2 \ln(emp)_{i,2017} + \beta_3 \Delta \ln(1+\tau_{jct}) \times \ln(emp)_{i,2017} + \eta_{ct} + \delta_j + \epsilon_{ijct}. \tag{2.12}$$

Panel B of Table 2.8 shows that the average effects of tariff increases remain significant even after controlling for firm size. Larger firms are less likely to adjust their import profiles. Specifically, they are less inclined to establish new product-country markets or terminate existing import relationships. However, as detailed later, overall, large firms account for a larger market share following a tariff increase.

More interestingly, the detailed information from LFTTD allows us to further study the source of extensive margin changes, to better understand reallocation across firms. In Appendix A.4, we further examine whether the extensive margin is induced from changes of the importing product margin or shifts in the sourcing country.

2.4.2 Comparing Starting/Ending with Continuing pairs

The last four columns in Table 2.1 indicates that the unit value difference between starting pairs and continuing pairs increases, while the unit value difference between continuing pairs and ending pairs decreases in response to tariff hikes. This section discusses the difference between starting (ending) and continuing pairs in terms of unit value, import quantity share and post-tariff import price, and explores how these differences depend on firm size.

First, we compare starting pairs with continuing ones by estimating

$$y_{ijct'} = \alpha + \beta_1 \Delta \ln(1+\tau_{jct}) + \beta_2 \mathbb{1}_{ijct}^{\text{new c-j}} + \beta_3 \Delta \ln(1+\tau_{jct}) \times \mathbb{1}_{ijct}^{\text{new c-j}} + \eta_{ct} + \delta_j + \epsilon_{ijct}, \quad (2.13)$$

where $\mathbb{1}_{ijct}^{\text{new c-j}}$ is the indicator that takes value 1 for starting pairs ijc at period t' compared to period t. $y_{ijct'}$ refers to firms' unit value, post-tariff import price and import quantity share at firm-product-country level of the firm at period t'. Standard errors are clustered at HS-8 digit level.

Panel A of Table 2.9 summarizes the estimation results. On average, firms with new relationships exhibit higher unit values and post-tariff import prices but a lower market share within each product-country market compared to continuing pairs. Among product-country pairs subject to tariff increases during the trade dispute, new relationships do not show significantly different unit values or post-tariff import prices. However, these new pairs capture a higher market share in affected markets compared to those in markets not significantly impacted by tariff increases.

Accounting for firm heterogeneity, new pairs established by larger firms are, on average, associated with higher import costs but smaller market shares compared to continuing pairs of large firms, as reflected in the interaction term $\mathbb{1}_{ijct}^{\text{new c-j}} \times \ln(emp)_{i,2017}$. In contrast, for affected product-country pairs, there are no significant heterogeneous effects by firm size when comparing new and continuing pairs.

The corresponding estimation for ending pairs are shown in Table 2.10. On average, ending pairs are associated with a higher unit value and post-tariff import price, constituting a lower market share than continuing pairs. However, for those product-country pairs that face higher tariffs during the U.S.-China trade dispute, even when the import cost is lower, a larger fraction of product-country pairs with low import costs are terminated during the period of increased tariff compared to those in less-affected markets. In terms of the heterogeneous effects, ending pairs of larger firms in the affected market have higher import cost compared to those of smaller firms.

Table 2.9: Comparing Starting and Continuing Pairs

	$ln(\text{unit value})_{ijct'}$	$ln(\text{quantity share})_{ijct'}$	$ln(\text{import price})_{ijct'}$
A. Average Effects			
$\mathbb{1}^{\text{new c-j}}_{ijct}$	0.299	-0.063	0.301
	(27.53)***	(46.29)***	(27.74)***
$\Delta\ln(1+\tau_{jct})$	-0.195	0.066	0.656
	(1.17)	(4.35)***	(3.95)***
$\mathbb{1}^{\text{new c-j}}_{ijct} \times \Delta\ln(1+\tau_{jct})$	0.024	0.118	0.019
	(0.23)	(7.31)***	(0.18)
Obs.	7858000	7858000	7858000
adjusted R^2	0.68	0.3	0.68
B. Heterogeneous Effects			
$\mathbb{1}^{\text{new c-j}}_{ijct}$	0.234	-0.053	0.24
	(13.06)***	(21.44)***	(13.41)***
$\Delta\ln(1+\tau_{jct})$	-0.315	0.045	0.552
	(0.94)	(0.98)	(1.65)*
$\mathbb{1}^{\text{new c-j}}_{ijct} \times \Delta\ln(1+\tau_{jct})$	-0.127	0.016	-0.054
	(0.43)	(0.26)	(0.18)
$\ln(emp)_{i,2017}$	0.021	0.001	0.022
	(8.59)***	(3.42)***	(8.98)***
$\Delta\ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	-0.04	0.007	-0.049
	(1.1)	(1.27)	(1.35)
$\mathbb{1}^{\text{new c-j}}_{ijct} \times \ln(emp)_{i,2017}$	0.019	-0.001	0.019
	(10.04)***	(4.44)***	(9.83)***
$\mathbb{1}^{\text{new c-j}}_{ijct} \times \Delta\ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	0.041	0.009	0.034
	(1.04)	(1.35)	(0.87)
Obs.	2464000	2464000	2464000
adjusted R^2	0.68	0.31	0.68

Notes: Panel A presents the estimation results of equation (2.13). Panel B further includes firm employment in 2017 as the measure of firm heterogeneity. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

2.4.3 Understanding Mechanisms Behind

The firm-product-country level regressions detailed above for starting and ending pairs indicate that larger firms tend to pay higher prices than smaller ones. In addition, starting pairs generally pay more than continuing pairs, whereas ending pairs pay more than continuing pairs, though less than starting pairs. Notably, for product-country pairs that undergo a higher tariff increase, the import of cheaper pairs is halted. Next, we will test the possible mechanisms as detailed below.

Is the heterogeneity in price driven by contract stickiness? If contract stickiness were a factor, we would expect that among products with higher contract stickiness, starting pairs would pay a significantly higher price than continuing pairs. To this end, we use the contract intensity measure from Nunn (2007) and utilize product-level variation to see if the starting pair-continuing pair unit value difference is more significant among products with

Table 2.10: Comparing Ending and Continuing Pairs

	ln(unit value)$_{ijct}$	ln(quantity share)$_{ijct}$	ln(import price)$_{ijct}$
A. Average Effects			
$\mathbb{1}_{ijct}^{\text{exit c-j}}$	0.282	-0.063	0.284
	(26.26)***	(47.54)***	(26.54)***
$\Delta\ln(1+\tau_{jct})$	0.124	-0.047	-0.091
	(1.31)	(3.38)***	(0.95)
$\mathbb{1}_{ijct}^{\text{exit c-j}} \times \Delta\ln(1+\tau_{jct})$	-0.544	0.188	-0.549
	(7.86)***	(19.47)***	(7.93)***
Obs.	16100000	16100000	16100000
adjusted R^2	0.68	0.34	0.68
B. Heterogeneous Effects			
$\mathbb{1}_{ijct}^{\text{exit c-j}}$	0.248	-0.058	0.254
	(13.37)***	(24.09)***	(13.72)***
$\Delta\ln(1+\tau_{jct})$	0.351	0.129	0.147
	(1.72)*	(4.87)***	(0.72)
$\mathbb{1}_{ijct}^{\text{exit c-j}} \times \Delta\ln(1+\tau_{jct})$	-0.649	0.137	-0.655
	(3.63)***	(5.45)***	(3.67)***
$\ln(emp)_{i,2017}$	0.024	0.001	0.025
	(10.32)***	(2.68)***	(10.71)***
$\Delta\ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	-0.105	-0.014	-0.107
	(4.04)***	(4.57)***	(4.13)***
$\mathbb{1}_{ijct}^{\text{exit c-j}} \times \ln(emp)_{i,2017}$	0.013	0	0.013
	(7.59)***	(1.12)	(7.37)***
$\mathbb{1}_{ijct}^{\text{exit c-j}} \times \Delta\ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	0.06	0	0.059
	(2.51)**	(0.14)	(2.43)**
Obs.	5067000	5067000	5067000
adjusted R^2	0.68	0.34	0.68

Notes: Panel A presents the estimation results of equation (2.13) for the ending pairs. Panel B further includes firm employment in 2017 as the measure of firm heterogeneity. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

higher contract intensity. Specifically, based on Table 2.9, we add contract intensity measure to do triple interaction. But contract stickiness cannot help us understand why ending pairs pay a higher price than continuing pairs.

Is the heterogeneity in price driven by variety switching/quality upgrading? If starting pairs choose varieties with higher quality or more attractive attributes that are more expensive than continuing pairs, we will see a more significant starting pair-continuing pair unit value difference. To this end, we use product differentiation index from Rauch (1999) to see if the starting pair-continuing pair unit value difference is more significant among differentiated products. However, if variety switching/quality upgrading story is working, it helps explain the starting pair-continuing pair unit value difference, but not the ending pair-continuing pair unit value difference.

Is the heterogeneity in price driven by cost saving? The cost saving story helps explain the ending pair-continuing pair unit value difference. As for starting pairs, even if they are more expensive than continuing pairs, they might not be more expensive than continuing pairs that firms already have.

As detailed later, we take the fact that larger firms pay higher prices than smaller ones as given, and use a model with a fixed import cost to explain other observations, while not excluding other possible explanations. That is, in the model, we are agnostic about why starting pairs pay a higher price than continuing pairs. Instead, we focus on how the size difference between firms leads to the unit value difference between starting pairs and continuing pairs, and its implications on tariff pass-through and welfare.

2.5 Summary

Using LFTTD data from the U.S. Census, we observe incomplete pass-through within stable trading relationships, where foreign exporters lower their export prices to offset tariff increases. Larger U.S. importers experience greater pass-through, bearing a higher share of the tariff burden. Tariff changes also induce responses along the extensive margin: newly established pairs pay higher import prices compared to continuing pairs. Moreover, new pairs, especially those of larger importers, capture a higher market share following tariff increases.

Our results are in line with previous literature. On the intensive margin, we find that the firm-level tariff pass-through is incomplete, aligning with the findings in Alviarez et al. (2023). On the extensive margin, we observe fewer new trading relationships and more dissolutions of old relationships, which is consistent with Handley et al. (2023), among others.

3 The Model

In this section we develop a model to understand the potential mechanisms behind the empirical findings. The model includes U.S. households, U.S. firms, and foreign exporters.

3.1 Setup

Each period t, U.S. firms draw a productivity level from some exogenous process. After the realization of productivity, U.S. firms make decisions. Therefore, the subscript t is omitted in the rest of the model.

Specifically, on the input side, U.S. firms decide whether to establish a product-country trading relationship (extensive margin) and determine the import quantity for each active relationship (intensive margin). The extensive margin trade-off arises from productivity gains from foreign inputs versus fixed import costs. On the intensive margin, import prices are determined by two-sided market power, which is shaped by bargaining power, demand and supply elasticities, and the relative importance of each trading partner in the other's overall trade. Consequently, the quantity and price in each firm-product-country relationship are simultaneously determined by both margins. On the output side, U.S. firms sell goods to U.S. households and other U.S. firms.

Production technology There is a unit mass of U.S. firms. A U.S. firm i hires labor L_i and purchases intermediate input X_i to produce y_i amounts of a *unique* variety i according to the following Cobb-Douglas production function:

$$y_i = A_i L_i^{1-\alpha} X_i^{\alpha}, \quad 0 < \alpha < 1, \tag{3.1}$$

where A_i denotes firm i's productivity. Intermediate inputs X_i are aggregated from domestic inputs h_i and foreign inputs m_i:

$$X_i = [h_i^{\frac{\sigma-1}{\sigma}} + m_i^{\frac{\sigma-1}{\sigma}}]^{\frac{\sigma}{\sigma-1}}, \quad \sigma > 1. \tag{3.2}$$

The domestic inputs and foreign inputs are aggregated from domestically produced and foreign-produced varieties, respectively, as follows:

$$h_i = \left(\int_0^1 h_{ik}^{\frac{\theta-1}{\theta}} dk \right)^{\frac{\theta}{\theta-1}}, \quad \theta > 1, \tag{3.3}$$

$$m_i = \left[\int_{b_i^*}^{\infty} (b_{ijc} \cdot m_{ijc})^{\frac{\epsilon-1}{\epsilon}} dF(b) \right]^{\frac{\epsilon}{\epsilon-1}}, \quad \epsilon > 1, \tag{3.4}$$

where j denotes the product, c represents the sourcing country and k is index for U.S. firms.[9] The foreign input variety is defined at the country-product level. θ governs the elasticity of substitution across domestic intermediates, ϵ governs the elasticity of substitution across foreign intermediates, while σ determines the elasticity of substitution between domestic and foreign intermediates. The term b_{ijc} signifies the demand shifter (or productivity shifter) for

[9] k can be equal to i, which forms a roundabout style production. U.S. firms use all the other U.S. firms' output as intermediate input, including the output from themselves.

product j produced by country c and exported to U.S. firm i, while b_i^* represents the shifter cutoff as detailed later. Firm i imports only the foreign varieties with demand/productivity shifters above this cutoff. The cumulative distribution function for the demand/productivity shifter is denoted by $F(b)$. Moreover, m_{ijc} denotes the amount of product j produced by country c and sold to firm i, and h_{ik} denotes the amount of U.S. firm k's output, which is used by firm i as an intermediate input. As in other importer models,[10] a more productive importer is generally a larger importer in terms of sales or import volumes.

Output market and domestic input of U.S. firms In the output market, U.S. firms engage in monopolistic competition. The output of U.S. firms is used to produce the final goods c_i for U.S. household consumption and as an intermediate input h_{ij} that is used by other U.S. firms:

$$y_i = c_i + \int_0^1 h_{ik} dk. \tag{3.5}$$

U.S. households. In each period, there is a representative household in the U.S. supply labor L inelastically given wage rate W to maximize the consumption of composite final good C:

$$C = \left(\int_i c_i^{\frac{\rho-1}{\rho}} di \right)^{\frac{\rho}{\rho-1}}, \tag{3.6}$$

subject to the flow budget constraint $PC = WL + \Pi$, where c_i represents the demand for variety i, $\rho > 1$ represents the elasticity of substitution across varieties i, each uniquely produced by one U.S. firm, Π denotes the profit of U.S. firms, $P = \left(\int_i {p_i}^{1-\rho} di \right)^{\frac{1}{1-\rho}}$ is the price of final good, and p_i is the price of variety i determined by firm i according to a standard constant markup pricing rule. The consumption maximization problem leads to a standard demand function:

$$c_i = \left(\frac{p_i}{P} \right)^{-\rho} C. \tag{3.7}$$

Roundabout production. In each period, a firm i sells output to other U.S. firms according to a standard demand function

$$h_{ik} = \int_0^1 \left(\frac{p_{ik}^H}{p_i^H} \right)^{-\theta} \left(\frac{p_i^H}{p_i^X} \right)^{-\sigma} X_i dk, \tag{3.8}$$

[10] For example, Gopinath and Neiman (2014), Halpern et al. (2015), Blaum et al. (2018, 2019), Lyu et al. (2023), Bernard et al. (2018), etc.

where p_{ik}^H is determined by firm i according to a standard constant markup pricing rule, $p_i^H = \left(\int_0^1 (p_{ik}^H)^{1-\theta} dk\right)^{\frac{1}{1-\theta}}$, $p_i^F = \left[\int_{b_i^*}^{\infty} \left(\frac{\tau_{jc} p_{ijc}^F}{b_{ijc}}\right)^{1-\epsilon} dF(b)\right]^{\frac{1}{1-\epsilon}}$, and $p_i^X = \left[(p_i^H)^{1-\sigma} + (p_i^F)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$.

Thus, we obtain a roundabout production structure in which firm output y_i serves both as a final consumption good and as an intermediate input in domestic production:

$$c_i + \int_0^1 h_{ik} dk = \left(\frac{p_i}{P}\right)^{-\rho} C + \int_0^1 \left(\frac{p_{ik}^H}{p_i^H}\right)^{-\theta} \left(\frac{p_i^H}{p_i^X}\right)^{-\sigma} X_i dk. \tag{3.9}$$

Throughout, without loss of generality, we assume $\rho = \theta$ so that price consistency is ensured between final consumer goods c_i and domestically produced intermediate inputs h_{ik}, such that $p_i = p_{ik}^H, \forall k$. Under this assumption, equation (3.9) can be rewritten as

$$y_i = c_i + \int_0^1 h_{ik} dk = \left(\frac{p_i}{P}\right)^{-\rho} C + \int_0^1 \left(\frac{p_i}{p_i^H}\right)^{-\rho} \left(\frac{p_i^H}{p_i^X}\right)^{-\sigma} X_i dk \equiv D p_i^{-\rho}, \tag{3.10}$$

where $D \equiv P^\rho C + (p_i^H)^\rho \left(\frac{p_i^H}{p_i^X}\right)^{-\sigma} X_i$ stands for the aggregate demand shifter.

Import price with two-sided market power The U.S. firm i's import price is determined by the marginal cost of production of the exporters and the market power of both the importers and exporters. There is only one exporter for each product-country level variety. The price that the firm i pays to the exporter is given as

$$\tau_{jc} p_{ijc}^F = \mu_{ijc} c_{ijc}(A_i), \tag{3.11}$$

where p_{ijc}^F is the pre-tariff import price (i.e., unit value),[11] μ_{ijc} is the markup, and $c_{ijc}(A_i)$ is the marginal cost of production of exporters. We let the marginal cost of production of exporters to be dependent on the importers' productivity, in specific, $c_{ijc}(A_i) = \tilde{\xi} A_i^{\tilde{\zeta}} + \tilde{\chi}_{jc}$. It enables the model to capture the fact that large importers often pay higher import prices than smaller importers, despite their greater market power, which would typically lead to

[11] In the model section and quantification exercise, we denote the pre-tariff import price (also referred to as the unit value or export price) as p_{ijc}^F, replacing the notation in the empirical part p_{ijc}. Moreover, the price for domestic intermediates is denoted as p_{ik}^H.

lower import prices.[12] The markup is given as

$$\mu_{ijc} = (1 - \iota_i)\mu_{ijc}^{\text{oligopoly}} + \iota_i \mu_{ijc}^{\text{oligopsony}}, \tag{3.12}$$

where $\mu_{ijck}^{\text{oligopoly}}$ denotes the exporters' market power, $\mu_{ijck}^{\text{oligopsony}}$ denotes the importers' market power, and $\iota_i \in [0, 1]$ represents a bargaining power that determines a relative importance of these two sides of market power. There is a variation in bargaining power among importers with $\iota_i = \hat{\xi}^{\iota} A_i^{\hat{\zeta}^{\iota}} + \hat{\chi}^{\iota}$. The variation not only adds additional flexibility to the model, but also proves to be crucial for explaining the heterogeneity in pass-through among importers, as detailed later.

The pricing function (3.12) is flexible and designed to be estimated using the data. It can be micro-founded by Nash-in-Nash bargaining as demonstrated in Alviarez et al. (2023). The import price is endogenous to the market shares of both importers and exporters, which depend on the importers' choice of import quantities.

On one hand, the exporters' market power is characterized as an oligopoly markup

$$\mu_{ijc}^{\text{oligopoly}} = \frac{\varepsilon_{ijc}}{\varepsilon_{ijc} - 1} \geq 1, \tag{3.13}$$

where $\varepsilon_{ijc} = (1 - s_{ijc})\epsilon + s_{ijc}\tilde{\nu}$ is importer's demand elasticity, $s_{ijc} \equiv \frac{\tau_{jc} p_{ijc}^F m_{ijc}}{\sum_{jc} \tau_{jc} p_{ijc}^F m_{ijc}}$ is the share of exporter's sales over importer i's total imports, with the tariff τ_{jc} of product j from sourcing country c, and ϵ is the elasticity of substitution between foreign intermediates. Also, $\tilde{\nu} = (1 - s_i^f)\sigma + s_i^f((1 - s_i^X) \times 1 + \rho)$, where $s_i^f = \frac{p_i^F m_i}{p_i^F m_i + p_i^H h_i}$ is the foreign input share among all intermediate inputs, and $s_i^X = \frac{p_i^X X_i}{p_i^X X_i + W L_i}$ is the intermediate input share among all inputs of the firm.[13] As in standard models of oligopoly markup (see, e.g., Atkeson and Burstein (2008)), the importer demand elasticity ε_{ijc} is decreasing in the exporters' sales share s_{ijc}, and the oligopoly markup $\mu_{ijc}^{\text{oligopoly}}$ is increasing in the exporters' sales share s_{ijc}.

On the other hand, the importers' market power is characterized as an oligopsony markdown

$$\mu_{ijc}^{\text{oligopsony}} = \kappa \frac{1 - (1 - x_{ijc})^{\frac{1}{\kappa}}}{x_{ijc}} \leq 1, \tag{3.14}$$

[12] The empirical results indicate that $c'_{ijck}(A_i) \geq 0$, i.e., larger U.S. importers purchase goods that require higher marginal cost of production. Blaum et al. (2019) documents that larger importers import higher quality goods that require higher marginal cost of production, which can be a possible micro-foundation for our empirical findings.

[13] If $\alpha = 1$ and $\sigma = 1$ in our model, the expression for $\tilde{\nu}$ is the same as Alviarez et al. (2023). ι_i is exogenously given in the model, motivated by the estimation in Alviarez et al. (2023).

where $\kappa \in (0, 1]$ is the returns to scale of exporters' production, $x_{ijc} \equiv \frac{m_{ijc}}{\sum_i m_{ijc}}$ is the share of importer i's quantity relative to the total quantity supplied by the exporter (j, c), and exporters' total cost of production is $TC_{ijc}(m_{ijc}) = \Phi_{ijc}(A_i)(m_{ijc})^{\frac{1}{\kappa}}$.[14] A direct implication is that the inverse export supply elasticity is $\frac{1-\kappa}{\kappa}$. Furthermore, if $\kappa = 1$, exporters have a constant returns to scale production function, and the importers' market power $\mu_{ijc}^{\text{oligopsony}}$ becomes 1, where there is no importer market power. Therefore, importer market power exists with $\mu_{ijc}^{\text{oligopsony}} < 1$ only if $\kappa \in (0, 1)$, which corresponds to exporters operating under a decreasing returns to scale production function. As in standard models of oligopsony markdown (see, e.g., Alviarez et al. (2023)), $\mu_{ijc}^{\text{oligopsony}}$ is decreasing in the importers' buyer share x_{ijc}.

It is straightforward that when importers' bargaining power is sufficiently large, meaning ι_i is close to 1, the import price mainly depends on the oligopsony markdown. Conversely, when importers' bargaining power is sufficiently small, meaning ι_i is close to 0, the import price primarily depends on the oligopoly markup.

Fixed import cost and tariff A U.S. firm i utilizes the entire continuum of domestic intermediate inputs but imports only a subset of foreign varieties due to the presence of fixed import costs, as in Gopinath and Neiman (2014). The fixed cost of importing, denominated in wage units, is represented by EN_i^{η}, where $E > 0$ captures the scale of cost, N_i represents the number of varieties imported by firm i, and $\eta > 0$ implies a convex cost in the number of varieties. Importing foreign varieties leads to both variety gains and efficiency gains if foreign varieties are more cost-effective than domestic ones. Consequently, a trade-off arises between the fixed import costs and the potential efficiency gains from accessing foreign varieties.

A firm faces tariff τ_{jc} when importing product j from country c. After tariff increase, firms are less likely to overcome fixed import cost, especially if the firm size is small.

Firm's optimization problem Each firm i chooses labor input L_i, output prices p_i and $\{p_{ik}^H\}$, and the set of domestic inputs $\{h_{ik}\}$, given the price of labor W and other firms' output price p_{ki}^H. They also choose the demand/productivity shifter cutoff b_i^* and the quantity $\{m_{ijc}\}$ for foreign intermediates, given the set of foreign intermediate input prices $\{p_{ijc}^F\}$.

The measure of imported intermediates is $N_i(b_i^*) = \int_{b_i^*}^{\infty} f(b)db$, where $f(b)$ is the fraction of foreign inputs with price adjusted demand/productivity shifter $\frac{b_{ijc}}{\tau_{jc} p_{ijc}}$ equals to b.[15] It is

[14] The exporters' marginal cost of production $MC_{ijc} \equiv c_{ijc} = \Phi_{ijc}(A_i)$, if $\kappa = 1$.

[15] $f(b)$ is the probability density function corresponding to the cumulative distribution function $F(b)$.

easy to see that this price-adjusted demand/productivity shifter serves as a sufficient statistic to represent the rank of gains from imported intermediates. Consequently, the number of imported intermediates is a function of this price-adjusted demand/productivity shifter.

The firm i solves the following profit maximization and cost minimization problem:

$$\pi(A_i) \equiv \max_{y_i, p_i, b_i^*} D^{\frac{1}{\rho}} y_i^{1-\frac{1}{\rho}} - \psi(A_i, b_i^*) y_i - (N_i(b_i^*))^\eta E, \tag{3.15}$$

where

$$\psi(A_i, b_i^*) \equiv \min_{L_i, h_{ik}, m_{ijc}} \left\{ W L_i + \sum_k p_{ik}^H h_{ik} + \int_{b_i^*}^{\infty} \tau_{jc} p_{ijc}^F m_{ijc} dF(b) \right\} \tag{3.16}$$

$$\text{s.t.} \quad y_i = 1, \quad (3.1), (3.2), (3.3), (3.4) \& (3.11) \quad \text{hold.} \tag{3.17}$$

Solving these problems leads to the following optimality conditions:

$$p_i = p_{ik}^H = \frac{\rho}{\rho - 1} \psi(A_i, b_i^*) \tag{3.18}$$

$$h_{ik} = \left(\frac{p_{ik}^H}{p_i^H} \right)^{-\theta} h_i \tag{3.19}$$

$$m_{ijc} = b_{ijc}^{\epsilon-1} \left(\frac{\tau_{jc} p_{ijc}^F}{p_i^F} \right)^{-\epsilon} m_i \tag{3.20}$$

$$h_i = \left(\frac{p_i^H}{p_i^X} \right)^{-\sigma} X_i \tag{3.21}$$

$$m_i = \left(\frac{p_i^F}{p_i^X} \right)^{-\sigma} X_i \tag{3.22}$$

$$X_i = \frac{\psi(A_i, b_i^*)\alpha}{p_i^X} y_i \tag{3.23}$$

$$L_i = \frac{\psi(A_i, b_i^*)(1-\alpha)}{W} y_i \tag{3.24}$$

$$y_i = \left(\frac{\psi(A_i, b_i^*)}{D^{\frac{1}{\rho}}} \frac{\rho}{\rho - 1} \right)^{-\rho} \tag{3.25}$$

$$\eta (N_i(b_i^*))^{\eta-1} E = \left(p_i^F \right)^{\epsilon} \left(\frac{\tau_{jc} p_{ijc}^F}{b_i^*} \right)^{1-\epsilon} \left(\frac{p_i^F}{p_i^X} \right)^{-\sigma} \frac{y_i}{(\epsilon - 1) A_i} \left(\frac{W}{1-\alpha} \right)^{1-\alpha} \left(\frac{p_i^X}{\alpha} \right)^{\alpha - 1}, \tag{3.26}$$

where $\psi(A_i, b_i^*) = \frac{1}{A_i} \left(\frac{W}{1-\alpha} \right)^{1-\alpha} \left(\frac{p_i^X}{\alpha} \right)^{\alpha}$, $p_i^H = \left(\int_0^1 (p_{ik}^H)^{1-\theta} dk \right)^{\frac{1}{1-\theta}}$, $p_i^F = \left[\int_{b_i^*}^{\infty} \left(\frac{\tau_{jc} p_{ijc}^F}{b_{ijc}} \right)^{1-\epsilon} dF(b) \right]^{\frac{1}{1-\epsilon}}$, and $p_i^X = \left[(p_i^H)^{1-\sigma} + (p_i^F)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}$.

3.2 Equilibrium

A competitive stationary equilibrium consists of prices $\{p_{it}, p_{ikt}^H, p_{ijct}^F, W_t, p_{it}^X, p_{it}^H, p_{it}^F\}_{i,k\in I, j\in J, c\in C}^{t\in[0,\infty)}$ and an allocation $\{C_t, c_{it}, y_{it}, X_{it}, L_{it}, m_{it}, h_{it}, h_{ikt}, m_{ijct}, b_{it}^*\}_{i,k\in[0,1], j\in J, c\in C}^{t\in[0,\infty)}$ such that for all t, (i) representative households choose C_t and c_{it} to solve their utility maximization problem; (ii) U.S. firms $i \in [0,1]$ solve the profit maximization problem and cost minimization problem by choosing output price p_{it}, p_{ikt}^H, labor input L_{it}, intermediate input X_{it}, the set of domestic inputs $\{h_{ikt}\}$, the demand/productivity shifter cutoff b_{it}^* and the quantity $\{m_{ijct}\}$ for foreign intermediates; (iii) labor market clearing condition holds, i.e., $W_t L = W_t \int_i L_i di$; (iv) goods market clearing condition holds, i.e., equation (3.9) holds.

3.3 Model Mechanism

In this section, we illustrate how tariff pass-through to U.S. importers varies based on their size, how an increase in tariffs asymmetrically affects the U.S. firms' import share, and how an increase in tariffs impacts the formation and dissolution of trading relationships.

For simplicity in exposition and to highlight the key mechanisms behind the empirical findings, we will discuss the scenario where all firms have identical bargaining power, i.e., $\iota_i = \iota$, and where more productive firms pay a higher price, i.e., $c'_{ijc}(A_i) > 0$.

3.3.1 Intensive Margin: Tariff Pass-Through and Import Share Adjustment

A full log-differential of the pricing equation (3.11) yields the log change in post-tariff import price, which is specified as follows:

$$d\ln(\tau_{jc} p_{ijc}^F) = d\ln\mu_{ijc} + d\ln c_{ijc} + d\ln\tau_{jc}, \tag{3.27}$$

where

$$d\ln\mu_{ijc} = \Gamma_{ijc}^s d\ln s_{ijc} + \Gamma_{ijc}^x d\ln x_{ijc}, \quad \Gamma_{ijc}^s \equiv \frac{\partial\ln\mu_{ijc}}{\partial\ln s_{ijc}}, \quad \Gamma_{ijc}^x \equiv \frac{\partial\ln\mu_{ijc}}{\partial\ln x_{ijc}}. \tag{3.28}$$

The pass-through elasticity of a tariff increase $d\ln\tau_{jc}$ is therefore

$$\Phi_{ijc} \equiv \frac{d\ln(\tau_{jc}p^F_{ijc})}{d\ln\tau_{jc}} = \Gamma^s_{ijc}\frac{d\ln s_{ijc}}{d\ln\tau_{jc}} + \Gamma^x_{ijc}\frac{d\ln x_{ijc}}{d\ln\tau_{jc}} + \frac{d\ln c_{ijc}}{d\ln\tau_{jc}} + 1. \tag{3.29}$$

It is straightforward to show how the exporter's sales share, the importer's quantity share, and the exporter's marginal cost of production respond to tariff changes:

$$\frac{d\ln s_{ijc}}{d\ln\tau_{jc}} = (1-\epsilon)(1-s_{ijc})\left[\frac{d\ln\tau_{jc}p^F_{ijc}}{d\ln\tau_{jc}} + \frac{d\ln\tau_{j'c'}\tilde{p}^F_{ij'c'}}{d\ln\tau_{jc}}\right], \tag{3.30}$$

$$\frac{d\ln x_{ijc}}{d\ln\tau_{jc}} = -(1-x_{ijc})\left[\varepsilon_{ijc}\frac{d\ln\tau_{jc}p^F_{ijc}}{d\ln\tau_{jc}} + \frac{d\ln\tilde{m}_{i'jc}}{d\ln\tau_{jc}}\right], \tag{3.31}$$

$$\frac{d\ln c_{ijc}}{d\ln\tau_{jc}} = \frac{1-\kappa}{\kappa}\left(-\varepsilon_{ijc}x_{ijc}\frac{d\ln\tau_{jc}p^F_{ijc}}{d\ln\tau_{jc}} + (1-x_{ijc})\frac{d\ln\tilde{m}_{i'jc}}{d\ln\tau_{jc}}\right), \tag{3.32}$$

where $\frac{d\ln\tau_{jc}\tilde{p}^F_{ij'c'}}{d\ln\tau_{jc}}$ represents the elasticity of the selling prices of an exporter (j,c)'s competitors in response to a tariff increase, and $\frac{d\ln\tilde{m}_{i'jc}}{d\ln\tau_{jc}}$ represents the elasticity of the quantities purchased by the competitors of importer i in response to a tariff increase.[16] Note that although we assume there is only one exporter for each product-country pair, each exporter treat other exporters as competitors, with the elasticity of substitution given by $\epsilon > 1$.

Therefore, under the assumption of competitors' non-responsiveness and sufficiently low values of each pass-through components, the import price pass-through elasticity is as follows:

$$\frac{d\ln(\tau_{jc}p^F_{ijc})}{d\ln\tau_{jc}} \approx \frac{1}{1+\Gamma^s_{ijc}(\epsilon-1)(1-s_{ijc})}\frac{1}{1+\Gamma^x_{ijc}\varepsilon_{ijc}(1-x_{ijc})}\frac{1}{1+\frac{1-\kappa}{\kappa}x_{ijc}\varepsilon_{ijc}} \tag{3.33}$$

$$\equiv \Phi^{SC}_{ijc}\Phi^{SS}_{ijc}\Phi^{TT}_{ijc}, \tag{3.34}$$

[16] The elasticity of the exporter's supplier share s_{ijc} with respect to the tariff increase, $\frac{d\ln s_{ijc}}{d\ln\tau_{jc}}$, can be derived in four steps. First, utilizing demand function of importers, we have $s_{ijc} = (\tau_{jc}p^F_{ijc}/p^F_i)^{1-\epsilon}$. Second, given the price index $p^F_i = (\sum_{jc}(\tau_{jc}p^F_{ijc})^{1-\epsilon})^{1/(1-\epsilon)}$, we have $\ln(p^F_i) = 1/(1-\epsilon)\ln(\sum_{jc}(\tau_{jc}p^F_{ijc})^{1-\epsilon})$, which leads to that $\partial\ln(p^F_i)/\partial\ln(\tau_{jc}p^F_{ijc}) = \partial\ln(p^F_i)/\partial(\tau_{jc}p^F_{ijc}) \times (\tau_{jc}p^F_{ijc}) = s_{ijc}$. Third, based on the second step, we have $d\ln(p^F_i) = \sum_{jc} s_{ijc}d\ln(\tau_{jc}p^F_{ijc}) = s_{ijc}d\ln(\tau_{jc}p^F_{ijc}) + (1-s_{ijc})d\ln(\tilde{p}^F_{ij'c'})$, where $d\ln(\tilde{p}^F_{ij'c'}) = \sum_{\forall j'c'\neq jc}\frac{s_{ij'c'}}{1-s_{ijc}}d\ln(\tau_{j'c'}p^F_{ij'c'})$. Fourth, based on the first and the third step, we expand $d\ln(s_{ijc})$ to derive the expression for $\frac{d\ln s_{ijc}}{d\ln\tau_{jc}}$. The elasticity of the importer's buyer share x_{ijc} with respect to the tariff increase, $\frac{d\ln x_{ijc}}{d\ln\tau_{jc}}$, is symmetrically derived in four steps. First, given the definition that $x_{ijc} = m_{ijc}/\sum_i m_{ijc}$, we have $\ln(x_{ijc}) = \ln(m_{ijc}) - \ln(\sum_i m_{ijc})$. Second, given that $\ln(m_{jc}) = \ln(\sum_i m_{ijc})$, we have $\partial\ln(m_{jc})/\partial\ln(m_{ijc}) = \partial\ln(m_{jc})/\partial m_{ijc} * m_{ijc} = x_{ijc}$. Third, based on step 2, we have $d\ln(m_{jc}) = \sum_i x_{ijc}d\ln(m_{ijc}) = x_{ijc}d\ln(m_{ijc}) + (1-x_{ijc})d\ln(\tilde{m}_{i'jc})$, where $d\ln(\tilde{m}_{i'jc}) = \sum_{\forall i'\neq i}\frac{x_{i'jc}}{1-x_{ijc}}d\ln(m_{i'jc})$. Fourth, Based on the first and third step, expand $d\ln(x_{ijc})$ to derive the expression for $\frac{d\ln x_{ijc}}{d\ln\tau_{jc}}$.

where

$$\Gamma^{s}_{ijc} = (1-\iota_i)\frac{\mu^{oligopoly}_{ijc}}{\mu_{ijc}}\frac{1}{\varepsilon_{ijc}-1}\frac{\epsilon-\varepsilon_{ijc}}{\varepsilon_{ijc}}; \tag{3.35}$$

$$\Gamma^{x}_{ijc} = \iota_i\frac{\mu^{oligopsony}_{ijc}}{\mu_{ijc}}\left[\frac{x_{ijc}(1-x_{ijc})^{\frac{1}{\kappa}-1}}{\kappa(1-(1-x_{ijc})^{\frac{1}{\kappa}})}-1\right]. \tag{3.36}$$

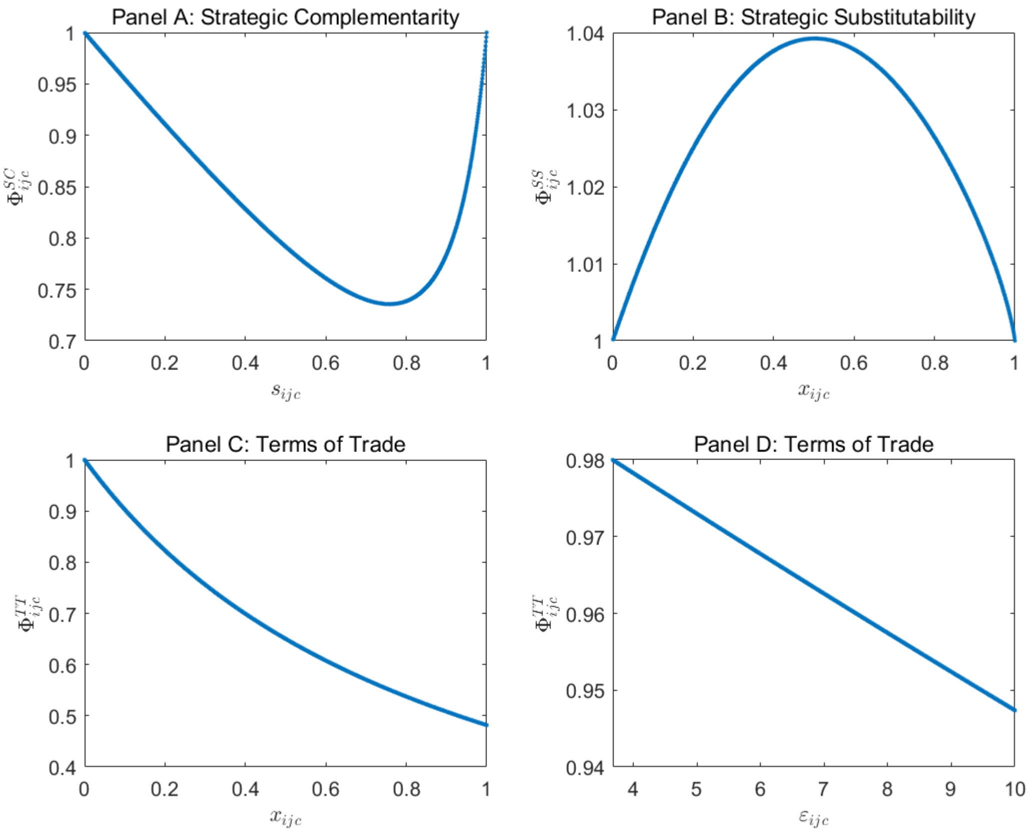
The tariff pass-through is therefore decomposed into three components. The first component $\Phi^{SC}_{ijc} \leq 1$ reflects strategic complementarities among exporters and highlights that exporters reduce their markup to stabilize their market share after tariff increases. Figure 3.1 panel A indicates that Φ^{SC}_{ijc} follows a "U-shape": when the exporters' supplier share is either very small or very large, there is high tariff pass-through to import price, implying minimal changes in markups. When $s_{ijc} = 0$, the exporter has no market power and behaves as a price taker, while the importer possesses significant market power over the exporter prior to the tariff increase. As a result, the exporter has no room to further reduce its markup in response to the tariff increase, leading to full tariff pass-through to the importer. In contrast, when $s_{ijc} = 1$, the exporter is sufficiently dominant to dictate prices to the importer, again resulting in complete tariff pass-through. However, when the buyer share is moderate, exporters are more willing to reduce their markups to maintain their market share, leading to lower tariff pass-through.

The second component $\Phi^{SS}_{ijc} \geq 1$ reflects strategic substitutability among importers, highlighting that importers raise import price by reducing markdowns following tariff increases. Figure 3.1 panel B indicates that Φ^{SS}_{ijc} follows a "hump-shape": when the importers' buyer share is either very small or very large, tariff pass-through to import prices is low, indicating less change in markdowns. However, when buyer share lies in the intermediate range, importers are more inclined to reduce markdowns, and hence face more tariff pass-through.

The third component $\Phi^{TT}_{ijc} \leq 1$ reflects the terms-of-trade channel. This component arises only when importers have market power $\kappa < 1$. Figure 3.1 panel C indicates a lower tariff pass-through to import prices as the importers' buyer share x_{ijc} increases, since larger importers adjust markdowns by more to cushion import price change. Panel D indicates a higher tariff pass-through to import price if there is a lower importers' demand elasticity ε_{ijc}, as importers with less elastic demand are more likely to absorb the tariff burden.

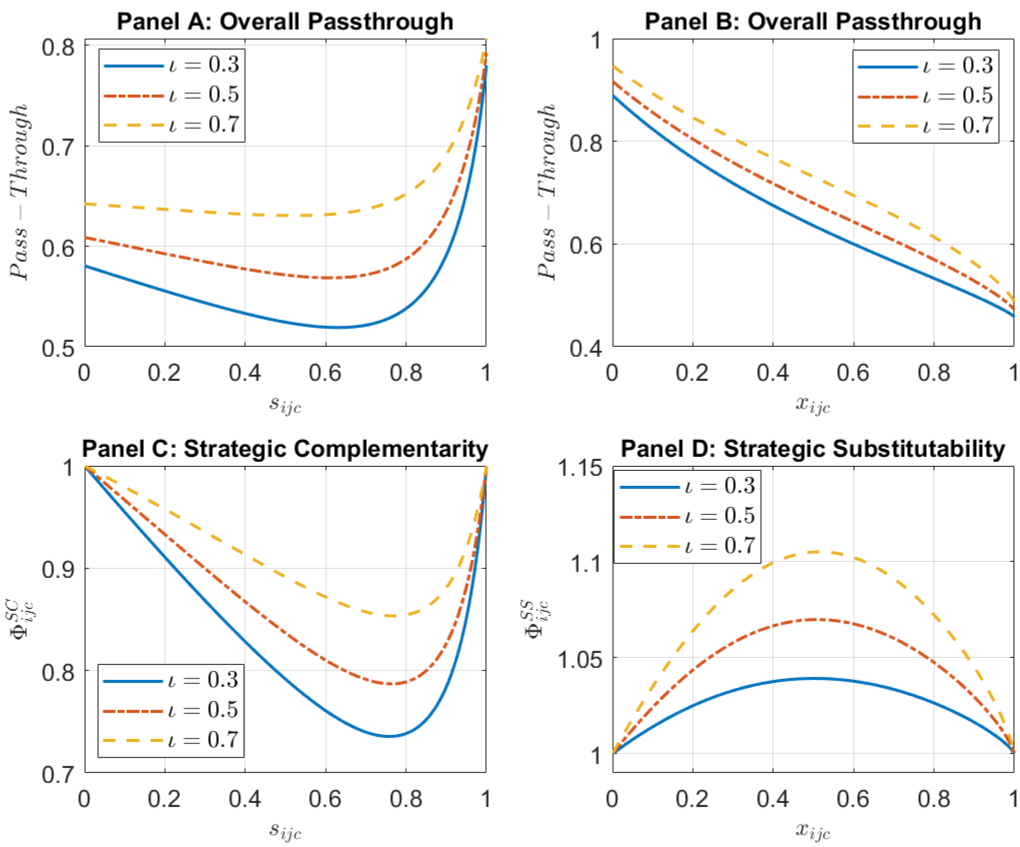
Importantly, the exporters' marginal cost of production $c_{ijc}(A_i)$ does not affect a firm's tariff pass-through for each imported variety, as it is independent of tariff change. Additionally, although our theoretical derivation of tariff pass-through makes assumptions on

Figure 3.1: Decomposition of Tariff Pass-Through Elasticities



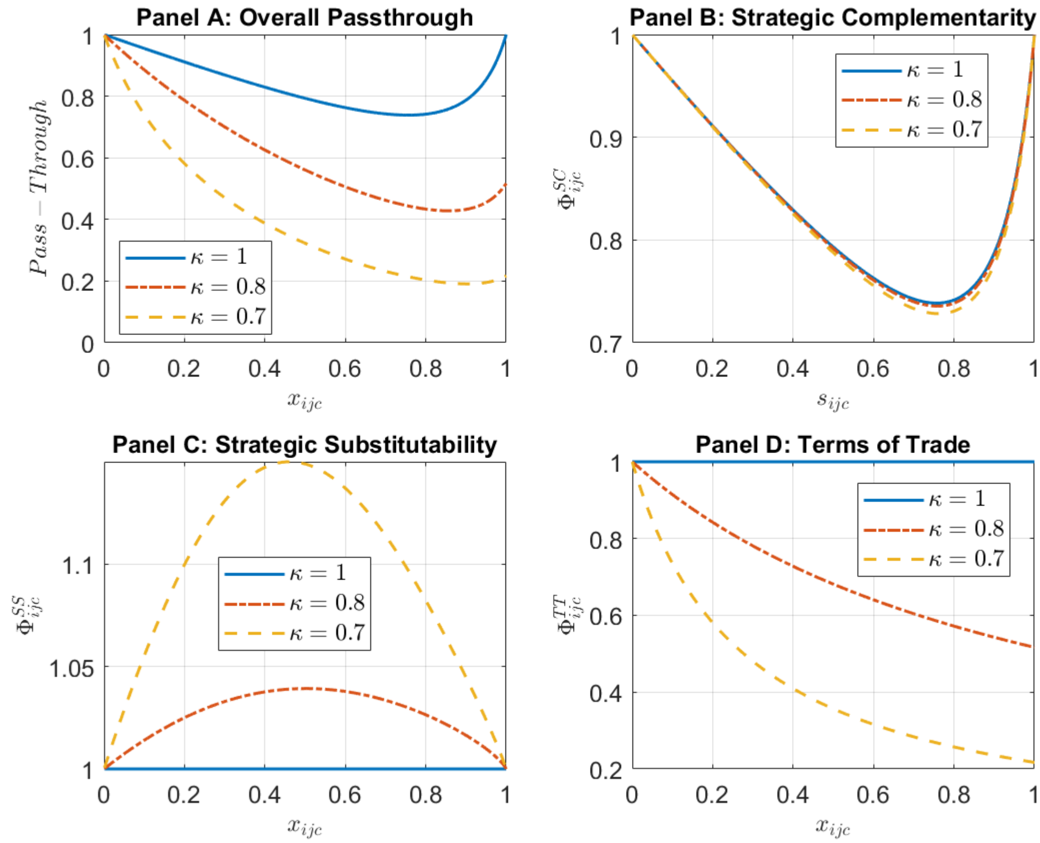
Notes: The figure illustrates how the three components of tariff pass-through elasticities—namely, the strategic complementarity component (SC), the strategic substitutability component (SS), and the terms of trade component (TT)—vary with model variables. In this plot, we fix importers' bargaining power $\iota = 0.3$, and importer market power $\kappa = 0.9$. Panel A shows that the SC component exhibits a "U-shape" relationship with exporters' supplier share s_{ijc}, defined as $s_{ijc} \equiv \frac{\tau_{jc} p^F_{ijc} m_{ijc}}{\sum_{jc} \tau_{jc} p^F_{ijc} m_{ijc}}$. Panel B shows that the SS component displays a "hump-shape" relationship with x_{ijc}, defined as $x_{ijc} \equiv \frac{m_{ijc}}{\sum_i m_{ijc}}$. Panel C shows that the TT component decreases as importers' buyer share x_{ijc} increases, while Panel D shows that the TT component decreases with importers' demand elasticity ε_{ijc}. A component value below 1 suggests incomplete tariff pass-through, a value above 1 implies more than complete pass-through, and a value equals to 1 indicates complete pass-through to import prices.

Figure 3.2: Change of Pass-through Components with Bargaining Power



Notes: The figure illustrates how the strategic complementarity component (SC), the strategic substitutability component (SS), and the overall tariff pass-through varies with the bargaining power ι_i, given $\kappa = 0.9$. The higher value of ι_i implies that the importer has relatively larger bargaining power over their exporters. Panel A shows that the pass-through increases with exporters' supplier share s_{ijc}. However, when importers' bargaining power is low, the pass-through exhibits non-linear relationship with supplier share. Panel B shows that the pass-through decreases with importers' buyer share x_{ijc}, and given buyer share, the tariff pass-through is more complete with higher importers' bargaining power. Panel C shows that the SC component exhibits a "U-shape" relationship with s_{ijc}, and importers face higher pass-through with higher bargaining power. Panel D shows that the SS component exhibits a "hump-shape" relationship with x_{ijc}, and the SS contributes more to complete pass-through as importers' bargaining power increase. Overall, given buyer share or supplier share, the tariff pass-through increases in importers' bargaining power.

Figure 3.3: Change of Pass-through Components with Importer Market Power



Notes: The figure illustrates how the strategic complementarity component (SC), the strategic substitutability component (SS), and the overall tariff pass-through-vary with importers' market power (exporters' return to scale) κ, given $\iota = 0.3$. Panel A shows that the overall tariff pass-through increases as importers' market power decreases. Panel B shows that the SC component is slightly affected by importers' market power, showing a modest increase in pass-through as importers' market power decreases. Panel C shows that the SS component rises significantly with higher importers' market power. Panel D shows that the TT component is absent when importers lack market power $\kappa = 1$. As importers' market power increases, the TT component shows a lower tariff pass-through.

competitors' market share adjustment, the numerical computations of the model without these assumptions confirm that the mechanisms align with the theoretical characterization of tariff pass-through.

As shown in Figure 3.2, three key findings emerge when studying the relationship between tariff pass-through and importers' bargaining power. First, in Panel A, a larger supplier share s_{ijc} tends to lead to higher tariff pass-through, but only when the importers' bargaining power ι is very high. In such cases, exporters have limited ability to reduce their markups, resulting in more direct transmission of tariff costs to import prices. However, when bargaining power is not very high, the relationship between supplier share and pass-through becomes non-linear. Notably, pass-through shows a "U-shape" pattern, where it is higher for very small or very large supplier shares but lower for intermediate levels. This non-linearity reflects exporters' strategic adjustments to maintain their market position under different levels of bargaining dynamics. Second, in Panel B, a larger buyer share x_{ijc} is associated with a lower tariff pass-through to import prices. This pattern arises because larger buyers have greater market power, allowing them to negotiate better terms and adjust markdowns more effectively to mitigate the impact of tariff changes. Third, holding all else constant, tariff pass-through increases with importers' bargaining power. When tariffs rise, the pre-tariff import prices of larger importers decline by less when importers have more bargaining power. This occurs because higher bargaining power limits exporters' ability to reduce markups in response to tariff shocks, as their pre-tariff prices are already suppressed through negotiations with strong importers. Consequently, importers bear a greater share of the tariff shock, amplifying the overall pass-through effect.[17]

Figure 3.3 illustrates how the strategic complementarity (SC) component, the strategic substitutability (SS) component, and the overall tariff pass-through change with importers' market power, represented by the exporters' returns to scale parameter, κ. Panel A shows that the overall tariff pass-through decreases as importers' market power increases. When importers have less market power (higher κ), they are in a weaker bargaining position relative to exporters, leading them to absorb a heavier tariff burden. Panel B shows that the SC component is slightly influenced by importers' market power, with a modest increase as importers' market power decreases. Panel C highlights the significant impact of market power on the SS component. The SS component is higher when importers have more market power (lower κ), reflecting that a more complete tariff pass-through. Panel D shows that TT component is absent when importers lack market power ($\kappa = 1$), but it becomes increas-

[17]Section B.2 provides more mdoel characterizations with comparative statics of ι and κ.

ingly relevant as market power increases. Higher market power ($\kappa < 1$) leads to a lower tariff pass-through. The underlying mechanism is that under constant returns to scale, a reduction in the exporter's production does not lower marginal production costs, generating a technological rent that can be negotiated between importers and exporters.

To summarize, the intensive margin adjustment highlights how tariff pass-through depends on exporter and importer market power as well as firm-specific characteristics. The decomposition into strategic complementarity (SC), strategic substitutability (SS), and terms-of-trade (TT) components reveals that SC component follows a "U-shape" relationship with exporters' supplier share, while the SS component exhibits a "hump-shape" pattern with importers' buyer share. Higher importers' bargaining power increases pass-through, as exporters have less flexibility to adjust their markups, shifting more of the tariff burden onto importers. Similarly, importers with less market power (higher κ) bear a larger share of the tariff burden, while those with greater market power see a higher SS component, leading to more complete pass-through. The TT component is only relevant when importers have market power ($\kappa < 1$). However, when the TT component is relevant, its effect is dominant, outweighing the influence of the SC and SS components and leading to an aggregate pattern where larger importers experience lower pass-through. Thus, to reconcile the discrepancies between macro and micro findings, larger importers must have stronger bargaining power, while importers as a whole must lack market power.

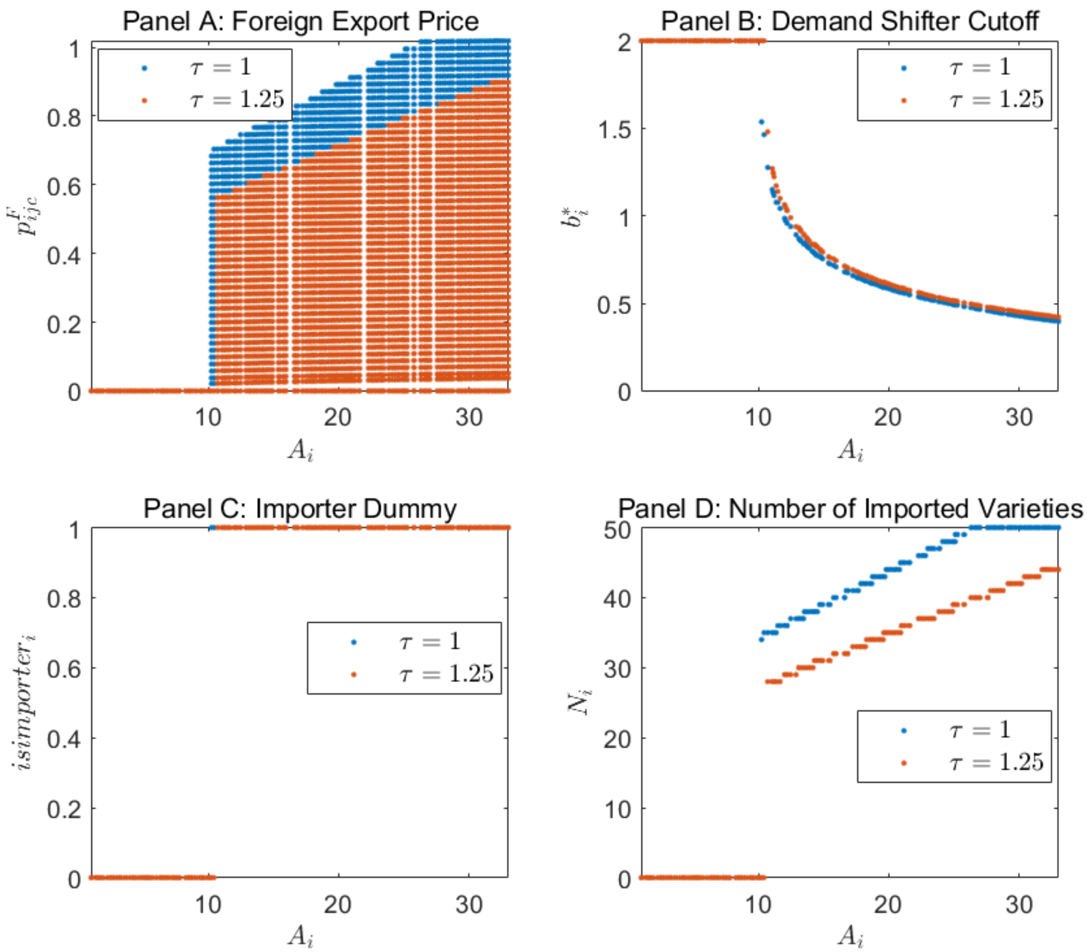
3.3.2 Extensive Margin: Tariff Increase and Trade Relationship Adjustment

Figure 3.4 provides a detailed illustration of how firms adjust their trade relationships extensively in response to a tariff increase of 0.25, shedding light on the impact of tariffs on foreign export prices, demand/productivity shifter cutoffs, import status, and the number of imported varieties.

Panel A reveals that more productive firms import intermediates with higher foreign export prices. This suggests that productive firms are willing to pay a premium for higher-quality or strategically important intermediates. However, when tariffs increase, the additional costs make these expensive intermediates less viable, forcing firms to eliminate such imports from their sourcing list.

Panel B illustrates that more productive firms have lower demand/productivity shifter cutoffs, indicating that they can sustain imports more easily due to their ability to overcome the convex import cost. The tariff hike, however, raises the cutoff for all firms, tightening

Figure 3.4: Extensive Margin Adjustment After Tariff Increases

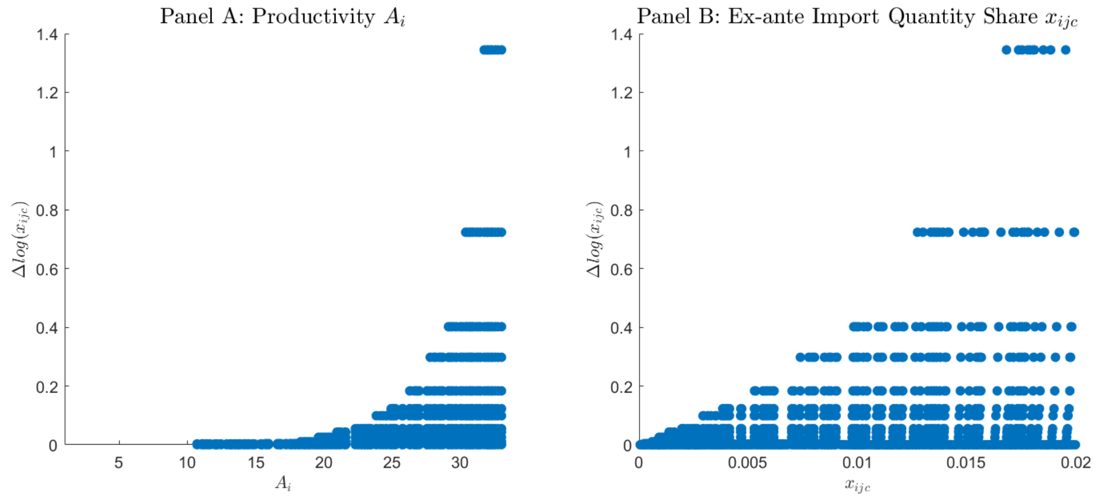


Notes: The figure illustrates how the foreign export price p^F_{ijc}, cutoff of the demand/productivity shifter b^*_i, import status, and number of imported varieties N_i-vary with firm productivity A_i before and after a uniform tariff increase of 0.25. Panel A shows that more productive firms import intermediates with higher foreign export price, and a tariff increase makes firms give up expensive foreign intermediates. Panel B shows that more productive firms have lower demand/productivity shifter cutoff, and a tariff increase raises the cutoff for all firms. Panel C shows that more productive firms are more likely to be an importer, and a tariff increase forces the marginal importers to exit. Panel D shows that more productive firms import more varieties, and a tariff hike decreases the number of varieties imported by all firms.

the criteria for import viability. This increase in the cutoff reflects a reduction in the attractiveness of foreign intermediates. The cost of introducing a marginal foreign intermediate outweighs the efficiency gain.

Panel C highlights that more productive firms are more likely to participate in international trade by importing intermediates. The tariff increase forces marginal importers to exit. Panel D shows that the number of imported varieties correlates positively with firm

Figure 3.5: Import Quantity Share Adjustment After Tariff Increases



Notes: The figure illustrates how the change in import quantity share $\Delta \log(x_{ijc})$, following a uniform tariff increase from $\tau = 1$ to $\tau = 1.25$, is affected by the heterogeneity of firms' productivity A_i, and their initial import quantity share x_{ijc}. Panel A shows that, after a tariff increase, imports become more concentrated among more productive firms. Panel B illustrates that imports are also more concentrated among firms with a higher initial import quantity share.

productivity, as more productive firms typically import a larger range of inputs to enhance the efficiency gains from foreign intermediates. After the tariff increase, all firms reduce their number of imported varieties.

Figure 3.5 illustrates how the change in import quantity share, $\Delta \log(x_{ijc})$, resulting from a tariff increase of 25%, is affected by the heterogeneity of firms' productivity A_i, and their initial import quantity share x_{ijc}. Panel A reveals that, following the tariff increase, imports become increasingly concentrated among more productive firms. This dynamic reflects a shift in trade relationships toward firms with higher production efficiency, as they are more capable of maintaining their access to foreign intermediates despite the tariff hike. Panel B reveals a similar concentration effect based on the initial import quantity share, x_{ijc}. This indicates that firms with an established position in international trade are more resilient to tariff-induced cost increases. However, firms with a smaller initial import share are disproportionately affected, resulting in a redistribution of import shares toward larger importers.

To sum up the extensive margin adjustment, the tariff increase leads to significant adjustments in firms' trade relationships, particularly in terms of their decision to import, and the range of intermediates they source. More productive firms demonstrate greater resilience, continuing to import despite the tariff hike, though with fewer varieties and higher thresholds for foreign intermediates. However, less productive firms are disproportionately affected, often exiting from importing. As a result, tariff increases lead to a contraction in trade relationships, reducing both the number of firms engaged in importing and the variety of intermediates sourced internationally. Trade is more concentrated among firms with higher productivity or higher initial trade presence. Since we assume that larger firms import foreign goods with higher marginal production costs, the extensive margin drives complete tariff pass-through at the product-country level.

4 Quantitative Analysis

As underscored by both the empirical findings and the model, various mechanisms are operating concurrently. In this section, we take the model to the data to determine the relative strength of each mechanism, interpret the empirical findings, and reassess the welfare implications of a tariff increase.

4.1 Parameterization

We categorize parameters as either externally or internally estimated. Table 4.1 lists all parameters and their estimated values.

4.1.1 External Estimation

The externally estimated parameters are those that govern the elasticity of substitution across output and inputs of importers and the size of tariff rates before the onset of the dispute. We set the share of material inputs in all inputs α to 1/2, a value which He (2024) calibrate to match the data in the U.S.[18] We set the elasticity of substitution across foreign varieties ε to be 2, following Alviarez et al. (2023), and the elasticity of substitution across domestic varieties θ to be 2 as well. We set the demand elasticity ρ that importers face to 2, which is similar to the U.S. downstream import demand elasticity in Broda and Weinstein (2006).The elasticity of substitution between domestic varieties and foreign varieties σ is set to 1.19. This value agrees with the estimates of substitution elasticity from Anderson and Van Wincoop (2004) and Edmond et al. (2023), who choose a similar value to match the

[18] Gopinath and Neiman (2014) finds $\alpha = 2/3$ using data in Argentina.

Table 4.1: Parameterization

Parameter	Notation	Value	Identification
Panel A. External Parameterization			
A1. Demand			
EOS across intermediate goods	ρ	2	Broda and Weinstein (2006)
A2. Production Technology			
Intermediate input share	α	1/2	He (2024)
EOS across domestic inputs	θ	2	Alviarez et al. (2023)
EOS across foreign inputs	ϵ	2	Alviarez et al. (2023)
EOS between H and F input	σ	1.19	Anderson and Van Wincoop (2004)
A3. Tariff			
Tariff before trade dispute	τ_{jc}	0.039	Average tariff across products
Panel B. Internal Parameterization			
Firm productivity	Scale	40.2	Firm import share, 9th decile
	Shape	62.19	Firm import share, 10th decile
	Persistence	0.31	Tariff increase
Demand/productivity shifter	γ	0.02	jc import share, 10th decile
Exporters' returns to scale	κ	0.99	P75 - P25: $\frac{d\ln(p_{ijc})}{d\ln(\tau_{jc})}$ of continuing pairs
Bargaining power $\iota_i(A_i)$	$\hat{\xi}$	0.01	Mean change in log p_{jc}
	$\hat{\zeta}$	0.37	Average component
	$\hat{\chi}$	0.01	Between component
Cost shifter $c_{ijc}(A_i)$	$\tilde{\xi}$	0.0001	Starting component
	$\tilde{\zeta}$	0.28	Starting share component
	$\tilde{\chi}_1$	0.45	Starting price component
	$\tilde{\chi}_2$	2.94	Ending component
Scale of import cost	E	0.01	Ending share component
Convexity of import cost	η	2.32	Ending price component

Notes: Unless otherwise specified, data moments are computed with LFTTD dataset for 2015–2017.

average markups in the U.S. These elasticity values are consistent with those in Fajgelbaum et al. (2020).[19] On the other hand, the tariffs prior to the U.S.-China trade dispute are computed using the LFTTD. Before the onset of the trade dispute, the average tariff rate was 3.9 percent.

4.1.2 Internal Estimation

The remaining internally estimated parameters are jointly determined to match the data moments that reflect the import distribution, tariff pass-through, and decomposition of ag-

[19] Fajgelbaum et al. (2020) estimates that in the U.S., the within product elasticity of substitution across countries is 2.53, the across products elasticity of substitution is 1.53, and the elasticity of substitution between imports and domestic products within a sector is 1.19.

gregate tariff pass-through, utilizing the simulated method of moments. Specifically, we simulate the model to represent two periods, before and after the trade dispute. We consider 10 percentage point increase in tariff rate and normalize the response. Without loss of generality, we suppose that there are no tariff variations across products nor countries. We simulate the model with 1000 firms, each potentially importing 10 products from 5 other countries. Then we compute the changes in the unit value between these two periods.

Import distribution U.S. firm productivity A_i is assumed to follow a Pareto distribution. We allow the firm-specific productivity to vary before and after the dispute to induce starting/ending a trading relationship for a product-country pair, with the persistence of ρ_A. Precisely, firms draw the initial productivity from the Pareto distribution with the scale A_x and the shape A_α. Then with the tariff increase, they draw a random shock ε_{Ai} so that their productivity then becomes $\rho_a A_i + \varepsilon_{Ai}$. The shock distribution of ε_{Ai} is such that the distribution across firms after the tariff increase remains to be Pareto with the scale A_x and the shape A_α. The estimated parameters for the productivity distribution are scale parameter of 40.26 and shape parameter of 2.19.

As for the functional form of the import demand/productivity shifter that governs the distribution of imports from different product-country pairs, we assume $b_{ijc} = \left((1-\gamma) + \gamma A_i^{\frac{\theta-1}{\theta}}\right)^{\frac{\theta}{\theta-1}}$ with $\gamma = 0.02$.[20] Therefore, the distribution across firms and that across imported varieties are both affected by the firm-level productivity. Both of these import share distributions are crucial as they impact the market share and, consequently, the market power of both U.S. importers and foreign exporters.

Size-dependent pass-through of continuing pairs The estimated exporters' returns to scale parameter κ is close to one (i.e., inverse foreign export supply elasticity is close to zero), and the bargaining power of importers ι_i increases with firm size, with $\hat{\zeta} > 0$. This is not surprising given our model mechanism discussion above. If κ is significantly smaller than one, the terms of trade channel will dominate and lead to larger firms facing less tariff pass-through. This is because, after a tariff increase, larger firms face a larger reduction in import demand, leading to a larger reduction in the marginal cost of exporters' production, and hence a larger reduction in export price and less pass-through for larger firms. However, this is inconsistent with the fact that larger firms that continue importing from a product-country pair face more tariff pass-through. On the other hand, even if

[20]We conduct sensitivity analysis for the parameter γ in Section 4.2.

κ is close to one, but ι_i is independent of firm size, the model predicts consistent pass-through across all importers who continue importing from the same product-country pair. In this scenario, the strategic substitutability channel dominates. Given that there is one exporter for each product-country pair and the exporters' markup is constant, all importers experience the same pass-through. When importers have larger bargaining power, it leads to lower markup adjustments from exporters. Therefore, ι_i increases with firm size helps to explain why larger importers experience more tariff pass-through. This result is supported by two pieces of evidence. First, as discussed in section 2.3.1, our empirical evidence suggests that the greater tariff pass-through faced by larger importers is not driven by the switching of varieties or exporters. Instead, it is related to factors that correlate with importer size. Second, previous empirical evidence, detailed in section 4.1.3, suggests that larger firms have higher bargaining power.

Pass-through via extensive margin response We set the marginal cost of production for exporters c_{ijc} to take the functional form of $c_{ijc}(A_i) = \tilde{\xi} A_i^{\tilde{\zeta}} + \tilde{\chi}_{jc}$ where A_i is the initial productivity of the firm i and $\tilde{\chi}_{jc} \sim U[\tilde{\chi}_1, \tilde{\chi}_2]$, and estimate the parameters $\tilde{\xi}, \tilde{\zeta}, \tilde{\chi}_1, \tilde{\chi}_2$.[21] The uniform distribution of c_{jc} is to allow for a variation across the variety. This reflects different costs of production across sourcing countries. We find that this marginal cost increases with firm productivity, with estimated $\tilde{\xi} > 0$ and $\tilde{\zeta} > 0$. This is not only consistent with the findings of Blaum et al. (2019), but also with our empirical evidence that larger firms face higher unit values, as shown in Table 2.9. According to the model specification discussed previously, the fact that larger U.S. importers encounter higher marginal costs from exporters is significant for the contribution of extensive margin adjustments by importers to aggregate tariff pass-through. Note that we do not take a stand on why larger importers face higher marginal production costs from exporters, but take it as given to examine how it matters for tariff pass-through. The previous literature (see, e.g., Blaum et al. (2019)) suggests that larger importers purchase higher-quality goods, which are inherently more expensive.

Targeted and untargeted moments The targeted moments from data and the model are presented in Table 4.2. The model closely generates qualitative results from the data. As in the empirical analysis, the pass-through at the jc level is (more than) complete in both data and the model. That is, with the increase in tariff, the unit price charged by exporters

[21] To better speak to the mechanism, we assume that there is no time variation in the cost c_{ijc} that arises due to firm-level productivity process over time. The result is similar even if we allow c_{ijc} to depend on the current period productivity and vary over time.

Table 4.2: Targeted Moments

Moment	Data	Model
A. Import Distribution		
Firm import share, 9th decile	0.01	0.00
Firm import share, 10th decile	0.99	1.00
jc import share, 10th decile	0.97	0.40
B. Size-Dependent Tariff Pass-Through of Continuing Pairs		
P75 - P25: $\frac{d\ln(p_{ijc})}{d\ln(\tau_{jc})}$ of continuing pairs	0.11	1.11
C. Aggregate Tariff Pass-Through		
Mean change in log p_{jc}	0.84	0.03
D. Decomposition of Aggregate Tariff Pass-Through		
Average	-0.25	-0.03
Between	-0.25	-0.02
Starting	1.82	2.74
Starting share	0.62	0.18
Starting price	4.53	2.74
Ending	-1.27	-7.14
Ending share	0.36	0.37
Ending price	-1.87	-7.08

Notes: The data moments in Panel A are computed from LFTTD, with details specified in Appendix B.3; the data moments in Panel B are from the regression results in Table 2.7; the data moments in Panels C and D are from the regression results in Table 2.1.

increase at the product-country level. However, once we exploit the firm-level data at the ijc level, the pass-through becomes negative. Additionally, we see that the decomposition result of the model qualitatively replicates the data. The ending component strongly pushes down the price changes, while average and between components have smaller but negative effects on the aggregate pass-through. Meanwhile, the starting component, especially its price rather than the share, leads to a larger pass-through. That is, in both data and the model, the jc level pass-through is driven by the increase in the unit price by the starting pairs.

Table 4.3 shows that the untargeted moments of the model qualitatively match the data, capturing the distributions of importer share, the share of imported varieties at the product-country level, and the sourcing strategies of firms. We compare the data prior to the U.S.-China trade dispute, specifically, from 2015 to 2017 with those from model simulation prior to the tariff increase.

Table 4.3: Untargeted Moments

Moment	Data	Model
A. Import Distribution		
Mean x_{ijc} of all pairs	0.13	0.03
Std x_{ijc} of all pairs	0.05	0.13
Mean s_{ijc} of all pairs	0.77	0.45
Std s_{ijc} of all pairs	0.13	0.36
B. Sourcing Strategy		
Average # imported varieties per firm at jc level	1.71	2.23
Std # imported varieties per firm at jc level	1.13	3.70
Import participation rate	0.06	0.40

Notes: The data moments are computed from the U.S. census 2015-2017 unless specified. Additional details on the construction of these moments can be found in Appendix B.3.

Summary As illustrated in the model, the import price depends on four factors: (i) the market share of both importers and exporters, (ii) the returns to scale for exporters, which are closely related to the foreign export supply elasticity, (iii) the relative bargaining power of importers, and (iv) the marginal cost of production for exporters. We target factor (i) to the data, estimate factors (ii) and (iii) by matching the size-dependent tariff pass-through of continuing pairs, and estimate factor (iv) by targeting the contribution of starting pairs to the aggregate tariff pass-through.

4.1.3 External Validation for Key Parameters κ and ι

While there might be concerns that these parameterization results for κ and ι heavily rely on the model setup, a substantial set of papers provides external validation. First, Fajgelbaum et al. (2020) estimates that the inverse foreign export supply elasticity is -0.002 and find they cannot reject horizontal foreign export supply curves. Fajgelbaum et al. (2024) estimates that US tariffs lead China to reduce exports to the US but also to increase exports to the rest of the world, and they cannot reject that Chinese (product-level) exports to the world remained constant. Jiao et al. (2023) uses firm-level data from one Chinese prefecture, and their estimates suggest no declines in firm-level sales to the world despite a decline to the US, which is again consistent with the relatively easy reallocation of Chinese producers across products and destinations.

Second, there is a body of research providing empirical evidence and mechanisms explaining why larger firms have higher bargaining power. Pfeffer and Salancik (1978), Porter

(1980), Sutton (1991), Dobson et al. (1998), Basker (2007), Mayer and Ottaviano (2008), and Ellison and Snyder (2010), among others, highlight that larger firms often have access to more resources and fewer dependencies. These may include greater financial capital, an established brand reputation, the ability to buy in bulk, switch suppliers with lower cost, or enter supply chain networks, which enable them to negotiate better terms with suppliers. Card et al. (2015) further substantiate this view by demonstrating how large firms dictate wage levels, work conditions, and even undermine competition.

4.2 Key Parameters in Pass-Through and Welfare Implication

In this section, we analyze the impact on the welfare of tariff increase. Welfare is measured by the change in consumption level due to the tariff increase in the baseline model. We also highlight how the key mechanisms, governed by various parameters of the model and firm heterogeneity, matter for the tariff pass-through at aggregate level and firm level, and thus for the welfare implication.

Welfare in baseline model Table **??** presents the model results. In the last row of column (1), we see that the tariff increase reduces the consumption. The reduction by 2.86 percent may appear large when compared to common values found in existing literature. This large value stems from our assumption of a uniform tariff increase of 10 percentage points across all products. As we are not precisely matching the tariff rate changes and other macro moments, this value should be regarded as a reference point for comparison with alternative models in our counterfactual analyses, rather than for comparing with welfare estimations in existing literature.

5 Conclusion

We use confidential firm-trade linked data from the U.S. Census to study the tariff pass-through during the U.S.-China trade dispute. Within continuing import relationships, foreign exporters partially accommodate tariff increases by lowering their export prices, resulting in incomplete pass-through to U.S. importers. The deviation from complete pass-through primarily stems from firm-level reallocation, a factor overlooked in previous studies focused on aggregate product-country level analyses. Specifically, two key patterns emerge. Within continuing pairs, import shares shift towards firms that experience higher pass-through. Second, starting pairs typically pay higher prices than continuing pairs. Besides, following the tariff increases, larger firms account for a greater share of imports and experience higher pass-through rates. Notably, larger firms also tend to pay higher prices than smaller firms.

Using an importer model that incorporates firm-specific import prices and two-sided market power, we show that an elastic foreign export supply, greater bargaining power of larger U.S. importers, and a fixed import cost that is independent of firm size, jointly rationalize the observed empirical patterns.

Our paper provides a wealth of empirical facts and theoretical mechanisms, enabling researchers to reevaluate the intensive and extensive margin adjustments of tariff pass-through during major trade policy changes. Additionally, our paper underscores the importance of firm-specific bargaining power in explaining pass-throughs in international trade. This mechanism, overlooked by previous literature, holds significant explanatory power for the high tariff pass-through to U.S. importers during the recent U.S.-China trade dispute.

It is intriguing to compare price responses across the short, medium, and long run to uncover the time-varying patterns of tariff pass-through. Our analysis focuses on the short run, as the detailed international transaction data is influenced by the subsequent COVID-19 pandemic in 2020. Exploring the long-run price dynamics following trade cost shocks remains an avenue for future research.

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A Data Appendix

A.1 Additional Tables

TBD.

A.2 Scheduled Tariff Construction

We generate a monthly time series representing U.S. scheduled import tariffs for the years 2018-2019 using data from Fajgelbaum et al. (2020). Their dataset offers origin-country-product level information on an annual basis for both 2018 and 2019. Within this dataset are details such as the effective month of tariff changes in 2018, maximum tariff, and scaled tariff. The authors obtained U.S. statutory import tariffs by scraping publicly available documents from the U.S. International Trade Commission (USITC). The USITC releases the benchmark tariff schedule annually, subsequently updating it throughout the unfolding trade war. These events of tariff adjustments cover the washing machine industry, solar panel industry, steel and aluminum industry, varieties with Chinese origin, and varieties with EU origin. Additional specifics can be found in their data appendix. Despite Fajgelbaum et al. (2020) providing a time series of scheduled tariffs, we opt to derive our own time series due to the original data being notably sparse, containing numerous missing values.

Our objective is to convert the dataset from an annual to a monthly basis. To achieve this, we employ the maximum tariff, scaled tariff, and effective month of tariff change to derive the monthly time series of U.S. scheduled import tariffs. The steps involved are as follows.

- **Step 1: Determining the Effective Month of Tariff Change in 2019.** We assume that there is only one tariff change per year for each product. Given that the dataset in their paper provides only the earliest change month of tariff, if there are subsequent tariff adjustments in 2019, we need to determine the effective change month for 2019. The scaled tariff denotes a weighted average of the cumulative tariff increases, with the weights reflecting the proportion of the year during which the tariffs were active.

$$\tau_{2019}^{Scaled} = \frac{M_{2019}^{E} - 1}{12}\tau_{2018}^{Max} + \frac{12 - M_{2019}^{E} + 1}{12}\tau_{2019}^{Max}$$

  where τ_{2019}^{Scaled} stands for the scaled tariff increase in 2019. M_{2019}^{E} denotes the effective change month of the tariff in 2019, indicating the month from which the tariff increase takes effect. For instance, if $M^E_{2019} = 9$, it signifies that from January to August, the tariff remains unchanged, while starting from September, the tariff is increased. τ^{Max}_{2019} denotes the maximum tariff increase in 2019. Therefore, we can rearrange the equation to derive M^E_{2019}.

$$M^E_{2019} = 12 \times \frac{\tau^{Max}_{2019} - \tau^{Scaled}_{2019}}{\tau^{Max}_{2019} - \tau^{Max}_{2018}} + 1$$

- **Step 2: Deriving the Initial Tariff Increase in 2018.** Again, utilizing the scaled tariff and the effective change month, we ascertain the initial tariff increase in January 2018. The scaled tariff increase in 2018 is derived as follows.

$$\tau^{Scaled}_{2018} = \frac{M^E_{2018} - 1}{12}\tau^{Initial}_{2018} + \frac{12 - M^E_{2018} + 1}{12}\tau^{Max}_{2018}$$

  where $\tau^{Initial}_{2018}$ is the initial tariff increase in 2018. After rearranging the equation, we can derive $\tau^{Initial}_{2018}$.

$$\tau^{Initial}_{2018} = 12 \times \frac{\tau^{Scaled}_{2018} - \tau^{Max}_{2018}}{M^E_{2018} - 1} + \tau^{Max}_{2018}$$

- **Step 3: Expand Annual data to Monthly data.** Generate year-month index. For example, 201901 denotes January 2019. Replicate $\tau^{Initial}_{2018}$, τ^{Max}_{2018}, τ^{Max}_{2019}, M^E_{2018}, and M^E_{2019} to monthly level on origin-country-product basis.

- **Step 4: Merge scheduled tariff in January 2018 from Fajgelbaum et al. (2020)** Obtain 201801 level scheduled tariff τ^{201801} from $m\_stattariff1$ in the replication file named $m\_flow\_hs10\_fm\_new.dta$. Then, merge the data series to the dataset generated from step 3 by year, month, hs10, and country. For the observations that miss τ^{201801}, we replace them to be 0. Till now, we have a time series for the monthly tariff increase from 2018 to 2019, as well as the scheduled tariff level of January 2018.

- **Step 5: Generate Monthly Time Series of Scheduled Tariff $\tau^{Scheduled}$.** If $Month < M^E_{2018}$, $\tau^{Scheduled} = \tau^{Initial}_{2018} + \tau^{201801}$. If $M^E_{2018} \leq Month < 201901$, $\tau^{Scheduled} = \tau^{Max}_{2018} + \tau^{201801}$. If $201901 \leq Month < M^E_{2019}$, $\tau^{Scheduled} = \tau^{Max}_{2018} + \tau^{201801}$. If $Month \geq M^E_{2019}$, $\tau^{Scheduled} = \tau^{Max}_{2019} + \tau^{201801}$. There are two special cases. First case is that M^E_{2018} is not existed. This means there is no tariff change in 2018. In this case, $\tau^{Scheduled} = \tau^{Max}_{2018} + \tau^{201801} = \tau^{Initial}_{2018} + \tau^{201801}$ for all months in 2018. Second case is that M^E_{2019} is not existed. This means there is no tariff change in 2019. In this case, $\tau^{Scheduled} = \tau^{Max}_{2018} + \tau^{201801}$ for all months in 2019.

A.3 Data Cleaning

Truncation issue and sample selection. Throughout, in all our regressions, for the unit value change and its four components, we drop outliers at top 5 percentile level and bottom 5 percentile level. For the change of tariff over time, we drop outliers at top 1 percentile level and bottom 1 percentile level. On the other hand, the entry and exit component are not correctly identified in the initial and end period of the data sample due to truncation issue. We therefore drop the truncation period to conduct our analysis.

A.4 New Product or New Country

In this section, we study whether the extensive margin changes arise from changes of the importing product margin or shifts in the sourcing country. If a firm has a starting import relationship ($\mathbb{1}_{ijct}^{\text{new c-j}} = 1$), they could fall into one of the four cases:

1. $\mathbb{1}_{ijct}^{\text{new j}} = 1$: firm i imports new product j from existing trading partner c at t'.
2. $\mathbb{1}_{ijct}^{\text{new c}} = 1$: firm i sources existing product j from a new trading partner c at t'.
3. $\mathbb{1}_{ijct}^{\text{new j, new c}} = 1$: firm i imports new product j from new trading partner c at t'.
4. $\mathbb{1}_{ijct}^{\text{old j, old c}} = 1$: Firm i imports existing product j from existing trading partner c at period t', but product j was imported from other existing countries at t.

As discussed in Section 2.4.1, firms, on average, are less likely to explore new product-country markets and more likely to terminate existing import relationships when tariffs increase. Larger firms experience less churn in their import profiles. These average and heterogeneous effects bury the variations about how firms create and dissolve product-country pairs.

To examine these dynamics in greater detail, we reestimate equations (2.11) and (2.12) by replacing the dependent variables with specific entry categories. The results, presented in Table A.1, show that firms are generally less likely to export to new markets or start importing new products, as indicated in Columns (1) to (3). However, firms tend to adjust their import strategies within existing trading relationships, as shown in Column (4).

This reshuffling of imported products within current trading partners remains the dominant pattern, when accounting for firm heterogeneity. Larger firms are more likely to explore

Table A.1: Probability of Starting

	(1) $\mathbb{1}_{ijct}^{\text{new c}}$	(2) $\mathbb{1}_{ijct}^{\text{new j}}$	(3) $\mathbb{1}_{ijct}^{\text{new j, new c}}$	(4) $\mathbb{1}_{ijct}^{\text{old j, old c}}$
A. Average Effects				
$\Delta \ln(1+\tau_{jct})$	-0.007	-0.137	-0.046	0.017
	(2.55)**	(18.59)***	(6.05)***	(3.83)***
Obs.	26110000	26110000	26110000	26110000
adjusted R^2	0.03	0.12	0.04	0.03
B. Heterogeneous Effects				
$\Delta \ln(1+\tau_{jct})$	-0.076	-0.164	-0.162	0.325
	(13.83)***	(7.48)***	(10.98)***	(22.88)***
$\ln(emp)_{i,2017}$	-0.002	-0.002	-0.016	0.014
	(28.56)***	(4.16)***	(63.13)***	(54.87)***
$\Delta \ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	0.009	0.009	0.017	-0.044
	(16.81)***	(3.38)***	(9.65)***	(23.08)***
Obs.	7833000	7833000	7833000	7833000
adjusted R^2	0.03	0.17	0.07	0.06

Notes: Panel A and B presents the estimation results of estimating equation (2.11) and (2.12), respectively. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

new trading partners and import new products but are less likely to adjust their import strategies within existing relationships.

Similarly, ending pairs ($\mathbb{1}_{ijct}^{\text{exit c-j}} = 1$) also have four alternatives:

1. $\mathbb{1}_{ijct}^{\text{exit j}} = 1$: firm i stops importing product j, but still imports other products from existing sourcing country c at period t'.

2. $\mathbb{1}_{ijct}^{\text{exit c}} = 1$: firm i stops sourcing from country c, but still imports product j from other countries at t'.

3. $\mathbb{1}_{ijct}^{\text{exit j, exit c}} = 1$: firm i stops importing product j, and stops importing from country c at t'.

4. $\mathbb{1}_{ijct}^{\text{sur j, sur c}} = 1$: firm i stops importing product j from country c, but product j is imported from other countries, and firm i still imports other products from country c at t'.

The estimation results are presented in Table A.2. On average, firms are more likely to terminate existing import relationships, either by ceasing to import specific products or by

Table A.2: Probability of Exiting

	(1) $\mathbb{1}_{ijct}^{\text{exit c}}$	(2) $\mathbb{1}_{ijct}^{\text{exit j}}$	(3) $\mathbb{1}_{ijct}^{\text{exit j, exit c}}$	(4) $\mathbb{1}_{ijct}^{\text{sur j, sur c}}$
A. Average Effects				
$\Delta \ln(1+\tau_{jct})$	0.014	0.126	0.091	0.084
	(6.07)***	(15.15)***	(13.93)***	(14.37)***
Obs.	26110000	26110000	26110000	26110000
adjusted R^2	0.03	0.04	0.03	0.03
B. Heterogeneous Effects				
$\Delta \ln(1+\tau_{jct})$	-0.017	0.186	-0.002	0.523
	(3.17)***	(9.02)***	(0.09)	(28.91)***
$\ln(emp)_{i,2017}$	-0.002	-0.004	-0.014	0.014
	(25.00)***	(13.05)***	(72.43)***	(57.00)***
$\Delta \ln(1+\tau_{jct}) \times \ln(emp)_{i,2017}$	0.003	-0.013	0.002	-0.05
	(5.06)***	(5.21)***	(1.02)	(25.37)***
Obs.	7833000	7833000	7833000	7833000
adjusted R^2	0.04	0.06	0.05	0.05

Notes: Panel A and B presents the estimation results of estimating equation (2.11) and (2.12), respectively. The difference terms are calculated as 12-month log difference. Standard errors are clustered at HS-8 digit level. t-statistics are reported in parentheses. * significant at 10%; ** significant at 5%; *** significant at 1%.

discontinuing imports from certain countries. Incorporating firm heterogeneity by size reveals that the dominance of firm-level adjustments along the product margin over the sourcing country margin persists even after accounting for firm size. Larger firms are less likely to terminate trade relationships at the product level than at the country level in response to tariff increases.

B Model Appendix

B.1 Computation Algorithm

1. Initialize the number of countries, products, and firms. Specifically, generate a state space with three-dimension matrix $nf \times nc \times np$, where nf denotes number of manufacturing firm, nc denotes country (excluding domestic country U.S. itself), and np denotes product.

2. Initialize parameters. Initialize the distribution of firm productivity $A_i \sim \hat{F}(a)$. Initialize unadjusted demand/productivity shifter b^u_{ijc}. The adjusted demand/productivity shifter is as follows: $b_{ijc} = b^u_{ijc}/(\tau_{jc}p^F_{ijc})$.

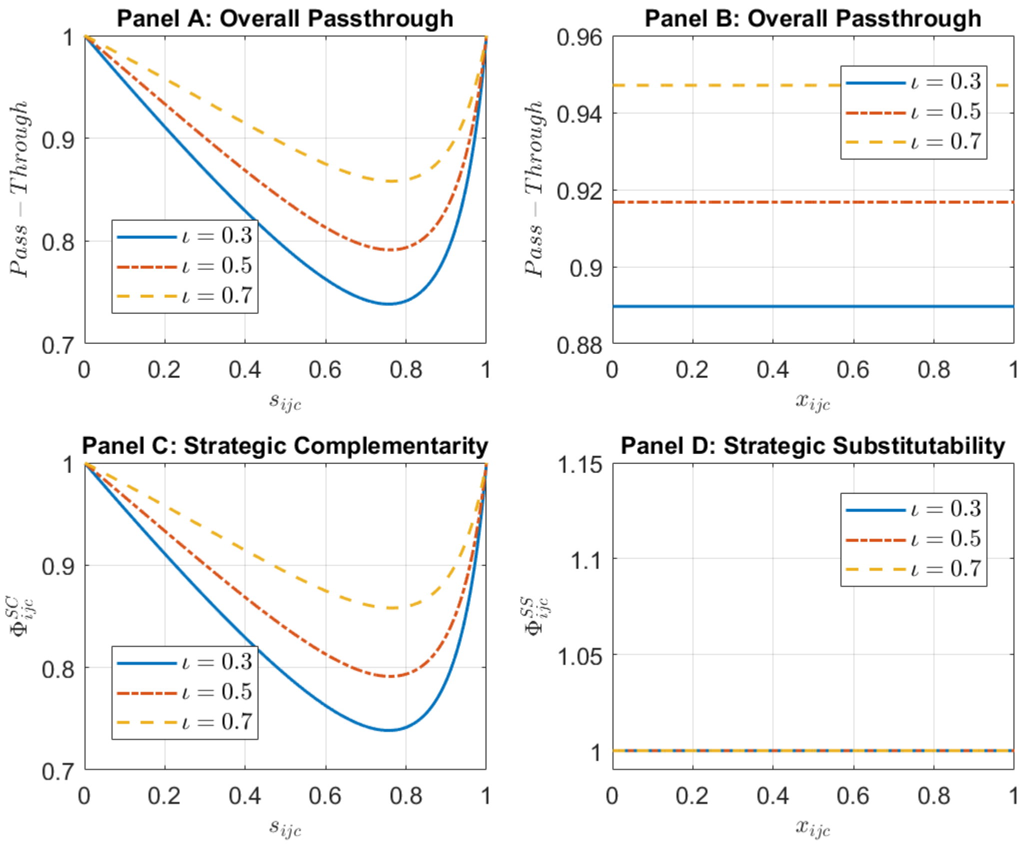
3. Inner loop: Guess the firm-specific cutoff threshold b^*_i for demand/productivity shifter. Firms will only import intermediates that is above this threshold, namely $b_{ijc} \geq b^*_i$. Given the cutoff, we solve the intensive margin adjustment by looping over the quantity of imported intermediates m_{ijc}, the quantity of domestic intermediates h_{ik}, the pre-tariff price of imported intermediates p^F_{ijc}, the price of domestic intermediate p^H_{ik}, demand shifter D, and wage W so that the criterion function converges to a value lower that $1e-6$. We update p^H_{ik} with the fact that the price for domestic intermediate is a constant markup over unit cost. We update D with $D = CP^\rho$, and we update W by imposing the labor market clearing condition.

4. Outer loop: Given the solved prices and quantities in inner loop, loop over demand/productivity shifter cutoff b^*_i so that the left hand side of equation (3.26) converges to a value lower that $1e-6$. We use binary search algorithm to clear the extensive margin condition. When left hand side of equation (3.26) is larger than 0, the marginal gain of including extra foreign intermediates outweighs the marginal import cost. In this case, we adjust the value of b^*_i down. When left hand side of equation (3.26) is smaller than 0, we update b^*_i with a higher value.

5. Going back to Step 4 (inner loop) with a new set of tariff τ_{jc}, and repeat the model simulation.

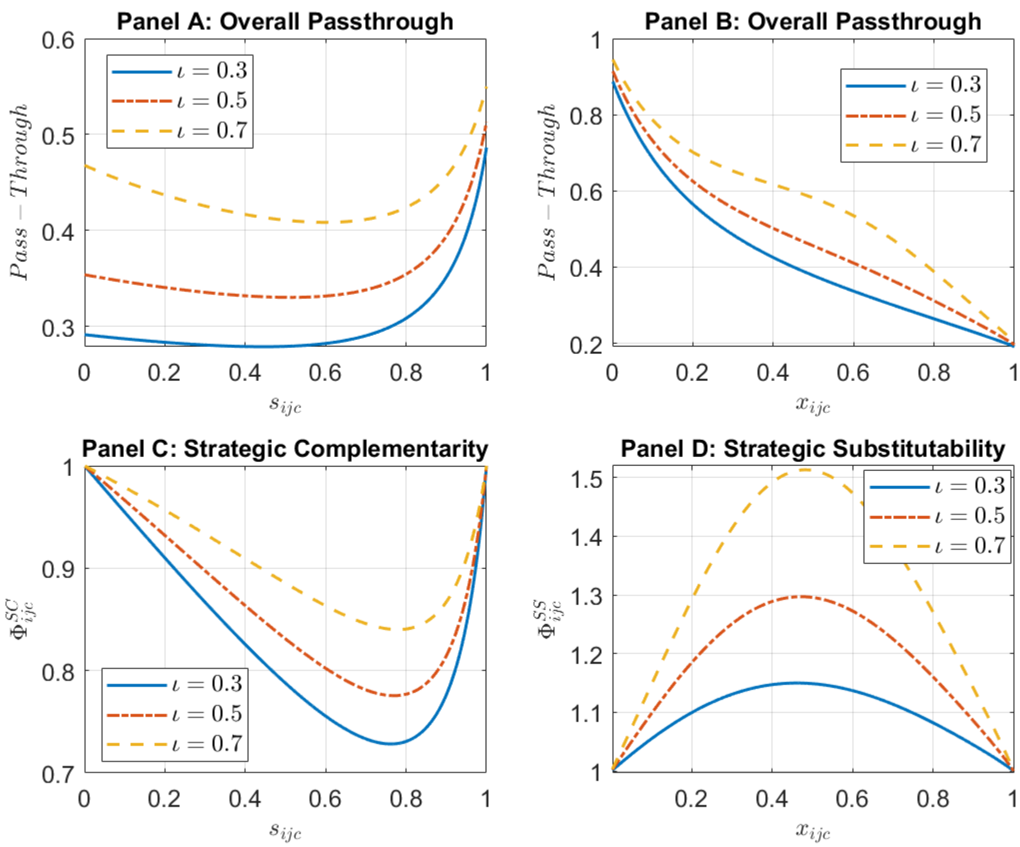
B.2 Additional Model Characterization

Figure B.3.1: ΔComponents with Importer Bargaining Power ι ($\kappa = 1$)



Notes: The figure illustrates how the strategic complementarity component (SC), the strategic substitutability component (SS), and the overall tariff pass-through-vary with importers' bargaining power ι, given $\kappa = 1$. Panel A shows that the pass-through increases with exporters' supplier share s_{ijc}. However, when importers' bargaining power is low, the pass-through exhibits non-linear relationship with supplier share. Panel B shows that when importers don't have market power, the pass-through is constant with importers' buyer share x_{ijc} when fixing supplier share s_{ijc}. Panel C shows that the SC component exhibits a "U-shape" relationship with supplier share s_{ijc}, and the overall pass-through is dominated by the SC component. Panel D shows that the SS component is constant given $\kappa = 1$. Also, Terms of Trade (TT) component is constant given κ. Thus, we don't plot it. Overall, the tariff pass-through increases in importers' bargaining power, and the pattern is governed by strategic complementarity component, when importer does not have market power.

Figure B.3.2: ΔComponents with Importer Bargaining Power ι ($\kappa = 0.7$)



Notes: The figure illustrates how the strategic complementarity component (SC), the strategic substitutability component (SS), and the overall tariff pass-through-vary with importers' bargaining power ι, given $\kappa = 0.7$. Panel A shows that the pass-through increases with exporters' supplier share s_{ijc}. However, when importers' bargaining power is low, the pass-through exhibits non-linear relationship with supplier share. Panel B shows that the pass-through decreases with importers' buyer share x_{ijc}. Panel C shows that the SC component exhibits a "U-shape" relationship with buyer share s_{ijc}, while Panel D shows that the SS component exhibits a "hump-shape" relationship with importers' buyer share x_{ijc}. Overall, the tariff pass-through increases in importers' bargaining power. There is more incomplete pass-through, when we increase importer market power (Comparing to Figure 3.2).

B.3 Targeted and Untargeted Moments

This section describes the procedure we take to compute moments from data that is to be compared with the model results.

1. ***Firm i's import shares***

We calculate each firm's total imports, denoted as M_{it}, over the period from 2015 to 2017. The import share for firm i is then defined as the ratio of its imports to the total imports across all firms, expressed as $M_{it}/\sum_i M_{it}$. We first take the mean and the standard deviation of firm i's import shares across time, and then target the average across firms.

Step 1: firm i import share during 2015-2017: $\frac{M_i}{\sum_i M_i}$, where $M_i = \sum_{jct} M_{ijct}$
Step 2: rank firms based on $\frac{M_i}{\sum_i M_i}$ (one firm one observation) and group them into 10 (4) groups. Calculate the group sum of $\frac{M_i}{\sum_i M_i}$ and group standard deviation of $\frac{M_i}{\sum_i M_i}$

2. ***Product-country jc's import shares***

We calculate the total imports for each product-country pair jc over the period 2015–2017, denoted as M_{jct}. The import share of a product-country pair jc is then defined as $M_{jct}/\sum_{jc} M_{jct}$. We target both the mean and the standard deviation of product-country jc's import shares.

Step 1: Product-country pair jc's import share during 2015-2017: $\frac{M_{jc}}{\sum_{jc} M_{jc}}$, where $M_{jc} = \sum_{it} M_{ijct}$
Step 2: rank jc based $\frac{M_{jc}}{\sum_{jc} M_{jc}}$ (one jc one observation) and group them into 10 groups. Calculate the group sum of $\frac{M_{jc}}{\sum_{jc} M_{jc}}$ and group standard deviation of $\frac{M_{jc}}{\sum_{jc} M_{jc}}$

3. ***Number of imported varieties*** N_i

For each firm i, we calculate the number of imported product-country varieties jc, denoted as N_{it}, over the period 2015–2017. We first calculate the average and standard deviation of the number of imported varieties within each firm, then target the average across firms.

Step 1: Firm i's number of imported varieties N_{it} during 2015-2017
Step 2: drop t dimension by calculating the mean and standard deviation of N_{it} across time for firm i, denoting as $\bar{N}_i$ and $sd_i(N_{it})$, respectively
Step 3: drop i dimension by taking the average of both $\bar{N}_i$) and $sd_i(N_{it})$ across firms.

4. ***Import participation rate***

We find Business Dynamics Statistics of U.S. Goods Traders reports import participation rate using data from LFTTD, so we report based on this public dataset.

5. x_{ijc} ***of all pairs***

The importers' quantity share is calculated as the ratio of importer i's import quantity in a product-country market jc at period t to the total import quantity in the same product-country market jc at period t. First, we compute the mean and standard deviate of the share for each firm i within a product-country pair over the period 2015–2017. Then, we calculate the average across firms.

Step 1: firm i quantity share at time t in jc market: $x_{ijct} = \frac{q_{ijct}}{\sum_i q_{ijct}}$ during 2015-2017
Step 2: drop t dimension by calculating the mean and standard deviation of $x_{ijct} = \frac{q_{ijct}}{\sum_i q_{ijct}}$ across time within a ijc pair
Step 3: drop jc dimension by calculating the average of (mean, sd) across jc within each firm i
Step 4: drop i dimension by calculating the average of (mean, sd) across i

6. s_{ijc} ***of all pairs***

The exporters' sales share is calculated as the ratio of importer i's one product-country pair's after-duty sales ($\tau_{jc} p^F_{ijc} m_{ijc}$) to importer i's total import value across all product-country pairs jc ($\sum_{jc} \tau_{jc} p^F_{ijc} m_{ijc}$) at period t. First, we compute the mean and standard deviate of the share for each product-country pair among all pairs within a firm over the period 2015–2017. Then, we calculate the average across firms.

Step 1: firm i sales share at time t in jc market: $s_{ijct} = \frac{sales_{ijct}}{\sum_i sales_{ijct}}$ during 2015-2017
Step 2: drop t dimension by calculating the mean and standard deviation of $s_{ijct} = \frac{sales_{ijct}}{\sum_i sales_{ijct}}$ across time within a ijc pair
Step 3: drop jc dimension by calculating the average of (mean, sd) across jc within each firm i
Step 4: drop i dimension by calculating the average of (mean, sd) across i

Step 1: firm i quantity share at time t in jc market: $x_{ijct} = \frac{q_{ijct}}{\sum_i q_{ijct}}$ during 2015-2017
Step 2: drop t dimension by calculating the mean of $x_{ijct} = \frac{q_{ijct}}{\sum_i q_{ijct}}$ across time within a ijc pair
Step 3: within a jc pair, rank firms based on their average import quantity share (already average across time), then calculate the ratio of average unit value in $> p75$ group to average unit value in < 25 group within that jc pair.
Step 4: drop jc dimension by calculating the average ratio

7. ***Tariff rate at*** jc ***market***

We compute the applied tariff τ_{jct} using data from the LFTTD for the years 2015–2017, and years 2018-2019 separately. For each product-country pair jc, we first calculate the time average. Then we target both the mean and the standard deviation of the applied tariff $\bar{\tau}_{jc}$.